GEMSTAR - TASK 19

LiDAR Processing, Sensor Observability, Adaptive Scheduling and ARM-FPGA Research

Final Illustrated Research Report

Based on recorded MATLAB and Python experiments, not on assumed hardware results.

Report generated: 2026-10-30 21:27 UTC

Research archive: Task_19_Final_Research

Evidence: 48 cross-recording frames; 47 evaluated frames; 286 original inventoried artifacts.

1. Executive Summary

Task 19 established a reproducible software baseline for GEMSTAR LiDAR preprocessing. It reconstructed OSI LiDAR reflections into 3D coordinates, tested ROI and voxel reduction, compared ground-plane estimation methods, examined clustering limitations, built sensor-observability and scheduling prototypes, and measured execution-time distributions.

Measurement	Result
Cross-recording observations	48
Evaluated processing frames	47
Selected-policy diagnostic failures	3
Median integrated speed ratio	1.398x
Median observability-monitor latency	0.15245 ms
Full baseline P95	3.41720 ms
Integrated policy P95	2.91280 ms
Illustrative deadline misses: full / policy	35 / 7

Interpretation: The adaptive software policy often reduced latency, but three evaluated configurations failed the experimental reference-consistency criteria. Sensor-observability warnings did not identify every processing-quality failure.

2. Research Workflow

Research step	Purpose / evidence
19.1-19.3	OSI reflection format, time-of-flight conversion, scan geometry, MATLAB XYZ visualization and internal geometry checks.
19.4-19.5	ROI filtering, provisional ground-height filtering and voxel centroid downsampling.
19.6	RANSAC ground estimation and comparison against simple ground-height classification.
19.7-19.8	DBSCAN clustering, bounding boxes, simulator object IDs, ground-truth inspection and world alignment diagnostics.
19.9	PC MATLAB preprocessing runtime and voxel-assisted RANSAC trade-offs.
19.10A-19.10J	Multi-frame reconstruction, offline oracle, rule-based scheduler, held-out testing and failure audit.
19.10K-19.10Q	Cross-recording validation, LiDAR observability, two-level policy analysis.
19.10R-19.10T	Feature overhead, integrated software processing and complete monitor timing.
19.11-19.12	Preprocessing consistency, deadline and tail-timing analysis.
19.13	Frame-150 INT8 and INT16 coordinate quantization experiment.

Research step	Purpose / evidence
19.14	Proposed ARM-FPGA partition and measurement plan.
19.15-19.16	Verified research consolidation, illustrated report and archive audit.

3. LiDAR Reconstruction

GEMSTAR produced 251 by 21 LiDAR reflection grids, corresponding to 5,271 rays per scan. Range was reconstructed from the recorded round-trip time of flight using the simulator's air light-speed constant. The ray-grid indices were mapped to horizontal and vertical scan angles, yielding sensor-local XYZ coordinates.

The reconstructed frame-150 candidate cloud contained 2,044 valid points. The original 20 m by +/-6 m ROI contained 1,922 points. Coordinate reconstruction was checked against the simulator source and its ray ordering, but independently verified object semantics were not established.

4. Geometric Preprocessing

Task 19 compared preliminary height-band ground filtering with an approximately horizontal RANSAC plane. It also studied 0.10 m, 0.25 m and 0.50 m centroid voxel grids as mechanisms to reduce plane-fitting workload. The full-resolution RANSAC output is a consistency reference, not a verified ground-truth segmentation.

5. Clustering and Object IDs

Initial DBSCAN clustering found two clusters in the frame-150 RANSAC non-plane points. Their geometry and simulator object-ID analysis revealed predominantly planar or elongated returns, not independently validated vehicles. This is an important negative finding: clustering cannot be equated with successful obstacle detection without semantic validation.

6. Scheduler Evaluation

An initial ROI-count rule selected between full-resolution and 0.25 m voxel-assisted RANSAC. Its original eight-frame agreement with an offline diagnostic oracle did not generalize perfectly to held-out frames. The rule was therefore preserved as a fixed baseline for subsequent recordings.

Across 48 cross-recording observations, the integrated processing policy executed 22 full-resolution paths, 25 voxel-assisted paths and one insufficient-data skip. Of 47 evaluated frames, three selected configurations failed the experimental processing-consistency criteria.

7. LiDAR Observability

Diagnostic state	Observations
OBSERVABLE	27
REVIEW	20
INSUFFICIENT	1

These states are provisional warning categories, not calibrated sensor-failure labels. Low return availability may arise from several causes. Monitoring return counts, range and spatial coverage does not by itself validate ground-segmentation accuracy.

8. Execution Time and Deadlines

The full monitor was benchmarked in MATLAB on a Windows PC. The recorded median monitor latency was 0.15245 ms. The overall processed-frame median baseline to integrated-policy latency ratio was 1.398x.

For an illustrative 3 ms computation budget, the recorded deadline violations changed from 35 baseline samples to 7 policy samples. The respective overall P95 values were 3.41720 ms and 2.91280 ms.

These figures are not real-time guarantees. They exclude OSI decoding, data transfer, sensor synchronization, downstream fusion and hardware power measurements.

9. Quantization Assessment

Bits	XYZ RMSE (mm)	Changed labels	Raw XYZ saving
8	43.404	4	87.5%
16	0.171	0	75.0%

The single-frame coordinate-quantization experiment favored INT16 for numerical precision. The study used frame-specific calibration and floating-point plane evaluation; it does not constitute an integer-only FPGA implementation.

10. ARM-FPGA Architecture

The proposed initial hardware kernel performs fixed-point point-to-plane distance calculation, inlier threshold comparison and inlier counting. The ARM CPU retains data management, scheduling, RANSAC hypothesis generation and initial SVD plane refinement.

The next hardware study must implement and verify RTL, integrate suitable streaming/DMA interfaces, and measure actual board latency, resource utilization and energy per frame. These measurements were not completed in Task 19.

11. Critical Research Limitations

- No independent semantic ground-truth accuracy for obstacle or road detection.

- Reference agreement may be high even when both algorithms are wrong.

- Only a small number of repeated PC timing samples were used.

- No measured FPGA LUT, FF, BRAM or DSP resources.

- No measured hardware power or energy savings.

- No full live camera-LiDAR-radar fusion.

- Quantization scales were assessed on one frame.

- Recordings may differ in more than weather.

- Low sensor return counts are not proof of failure.

- The current scheduling thresholds are exploratory.

12. Recommended Next Research

- Validate quantization on multiple recordings with fixed, independently calibrated scales.

- Implement an RTL fixed-point distance/inlier kernel.

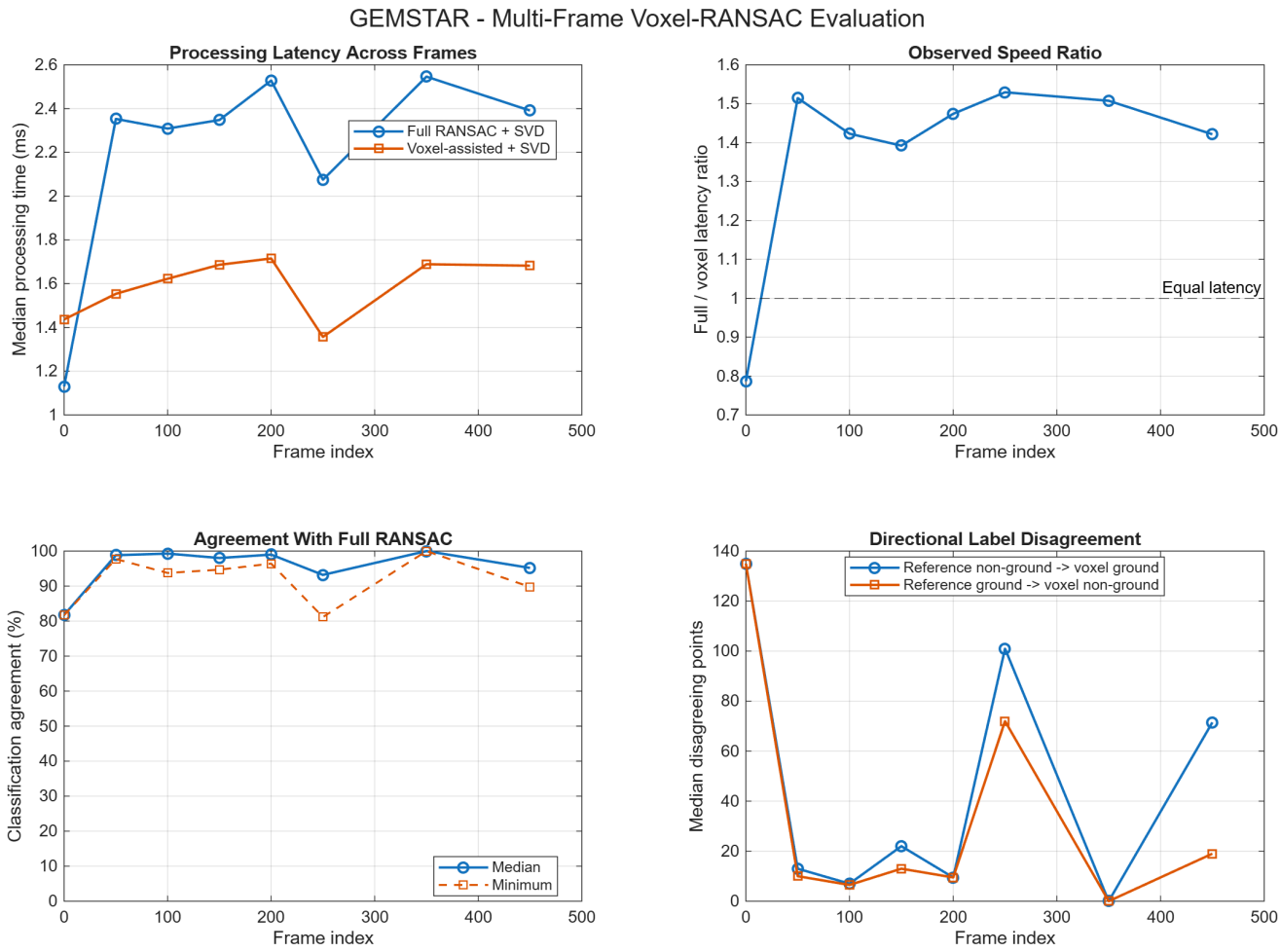
- Compare RTL and CPU results using identical inputs.
- Add ARM-FPGA data-transfer timing and buffering.
- Measure actual board power and energy per frame.
- Develop independent road/obstacle reference labels.
- Integrate camera and radar observability evidence.
- Evaluate the scheduler with deadlines and measured energy as explicit constraints.

13. Illustrated Experimental Evidence

The following pages reproduce the saved research figures from the Task 19 archive. Figure captions identify their source files. They are experimental evidence and should be interpreted using the limitations above.

Research Figure 1

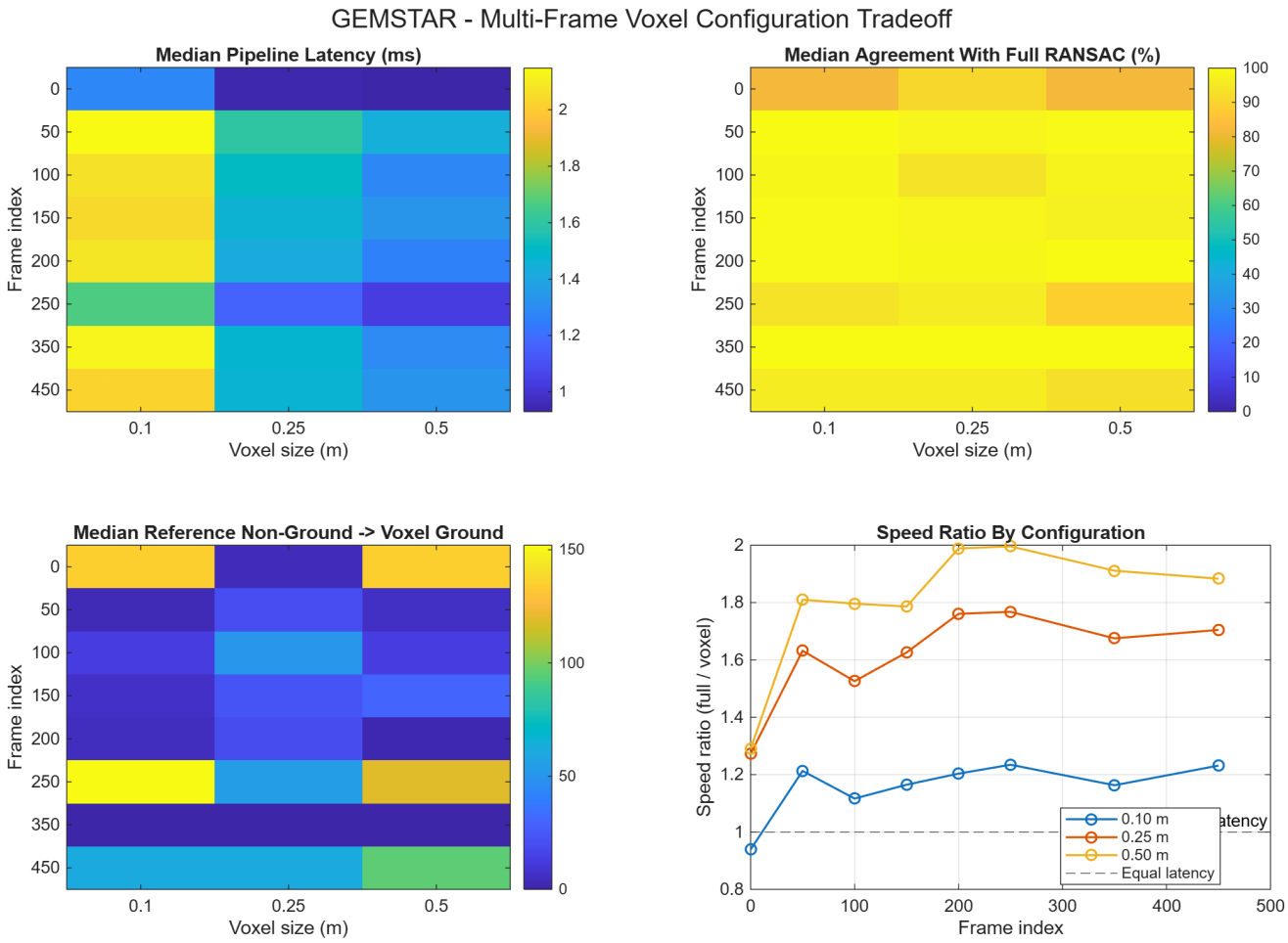
multiframe ransac benchmark



Original saved graph: 01_Graphs\task19_10c_multiframe_ransac_benchmark.png

Research Figure 2

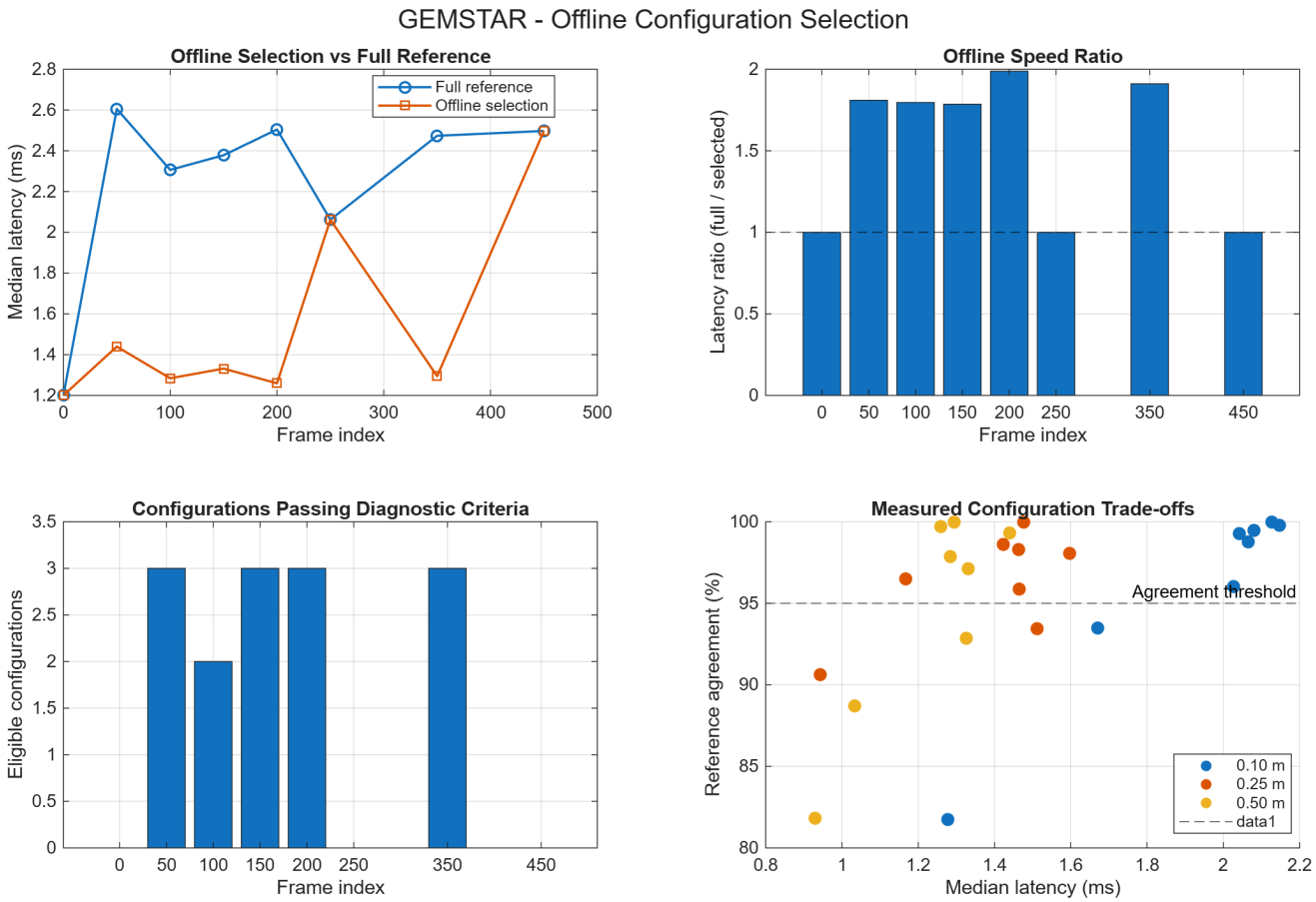
multiframe voxel tradeoff



Original saved graph: 01_Graphs\task19_10d_multiframe_voxel_tradeoff.png

Research Figure 3

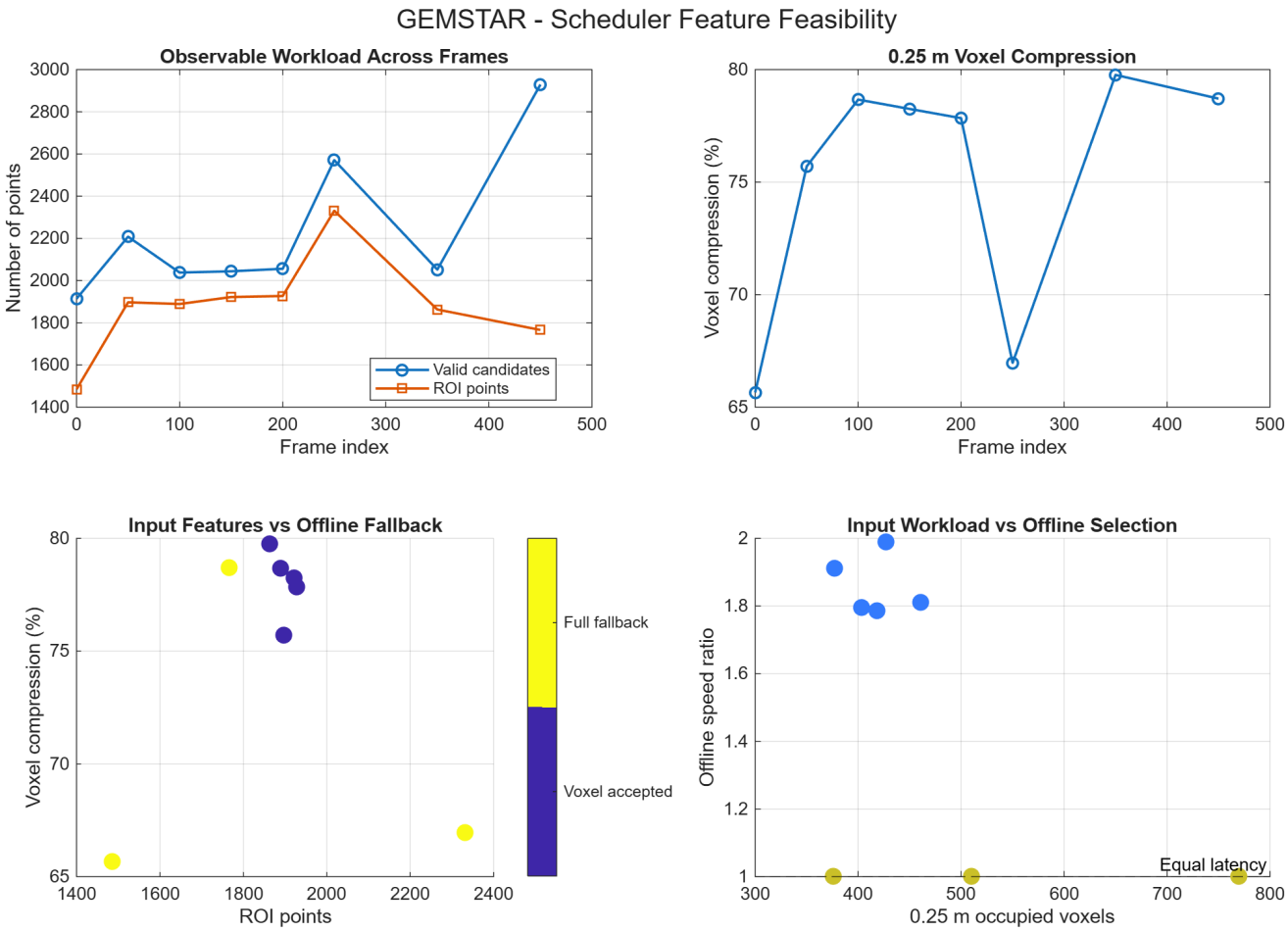
offline configuration analysis



Original saved graph: 01_Graphs\task19_10e_offline_configuration_analysis.png

Research Figure 4

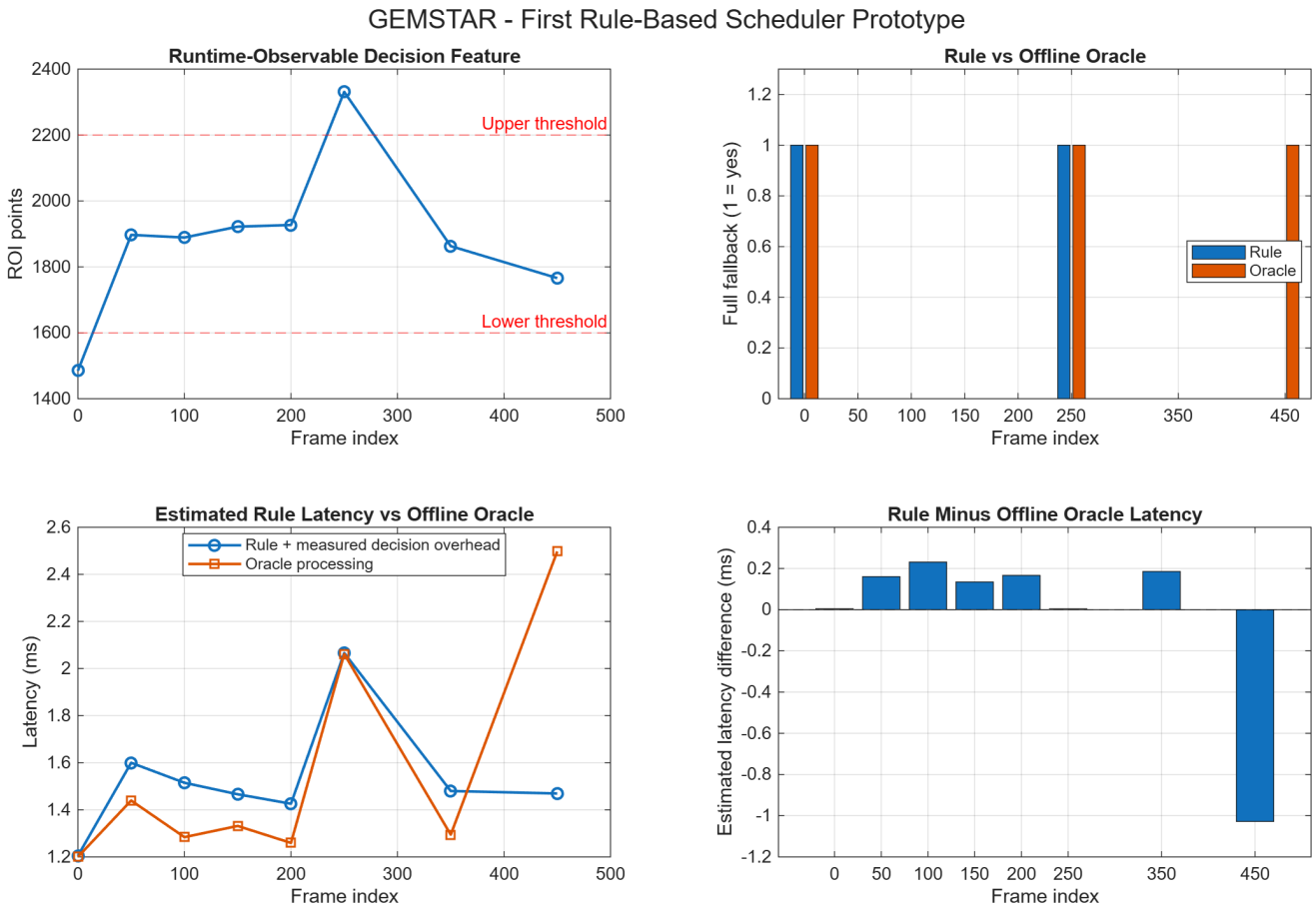
scheduler feature analysis



Original saved graph: 01_Graphs\task19_10f_scheduler_feature_analysis.png

Research Figure 5

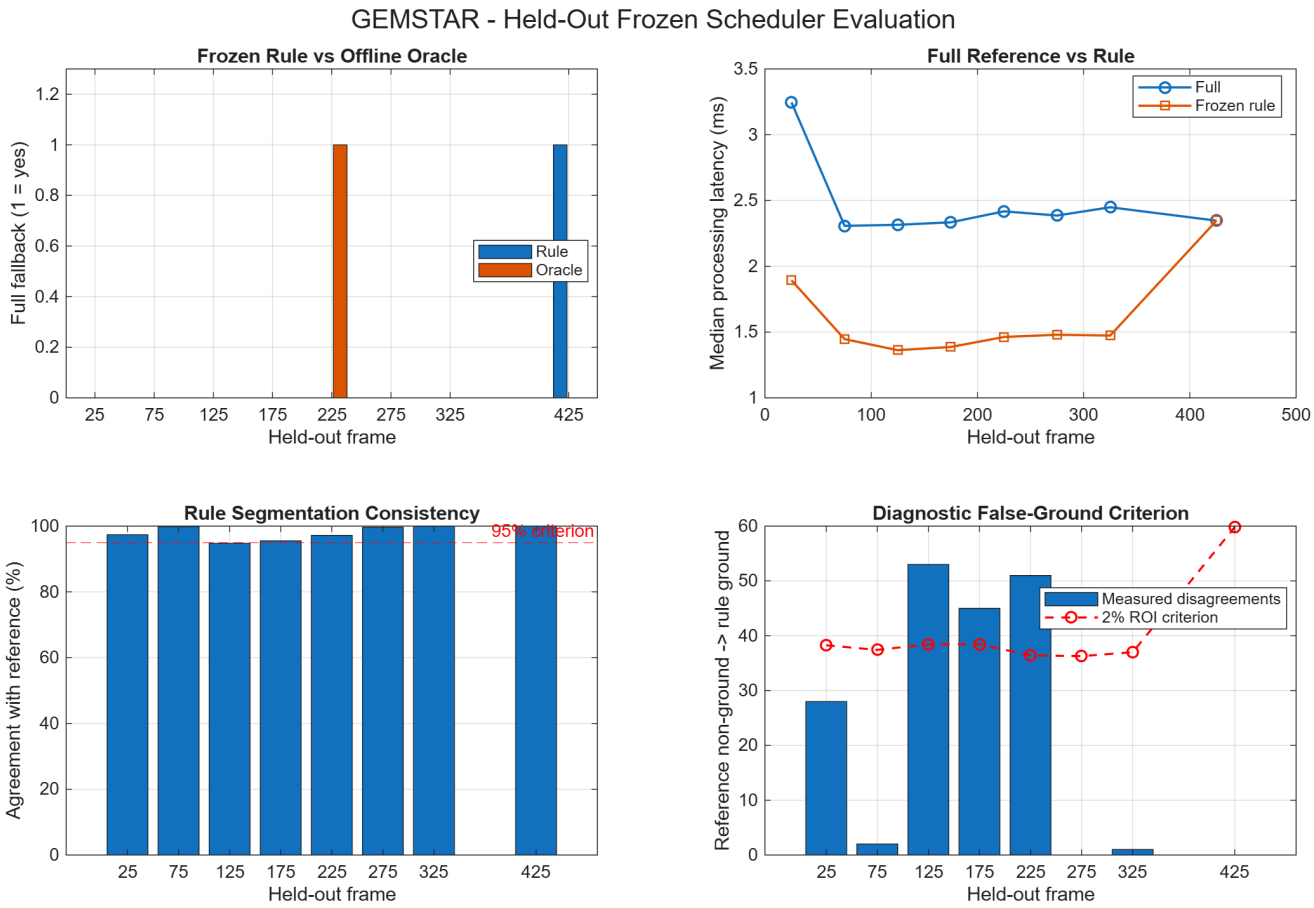
rule scheduler analysis



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Research Figure 6

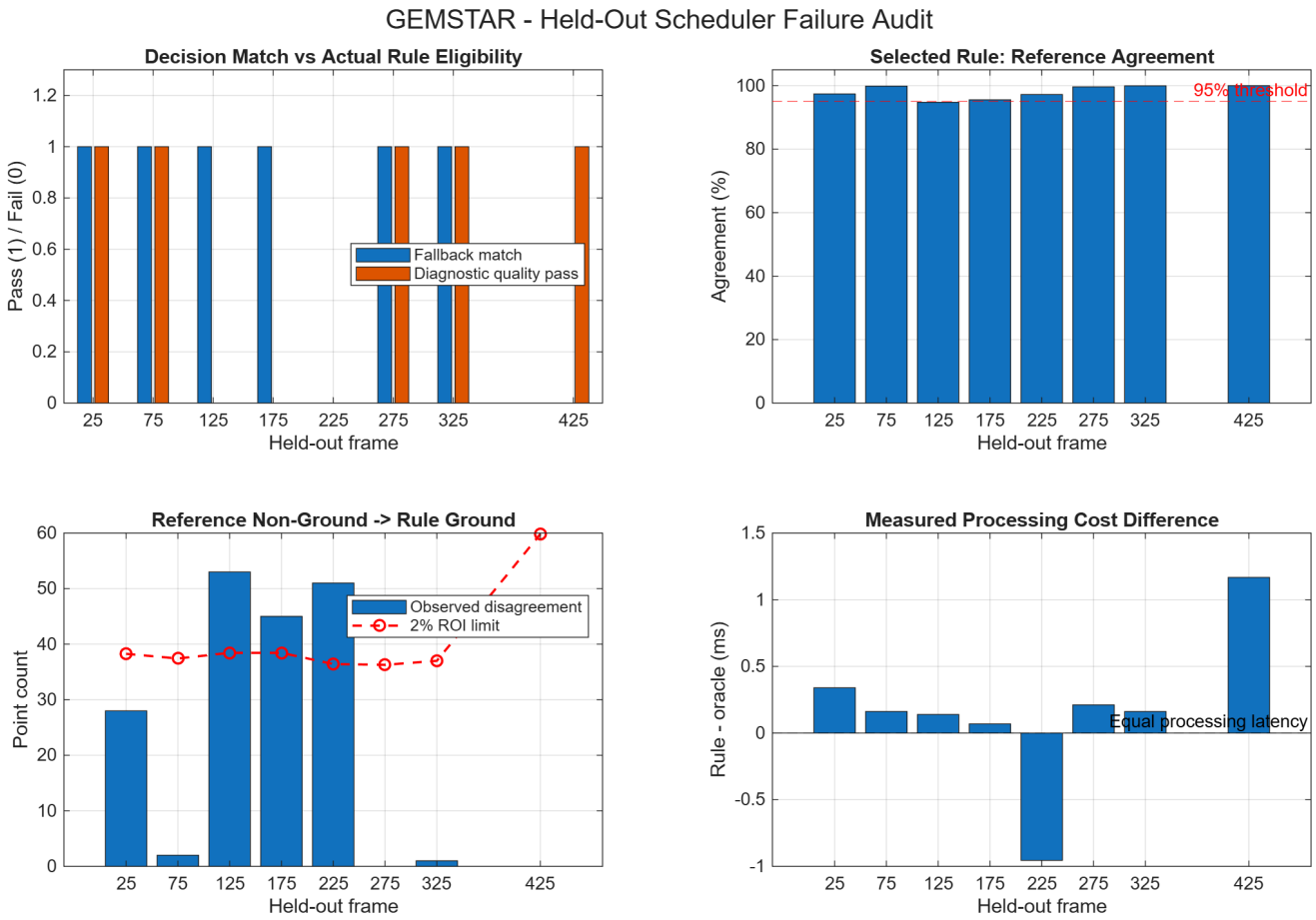
heldout scheduler evaluation



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Research Figure 7

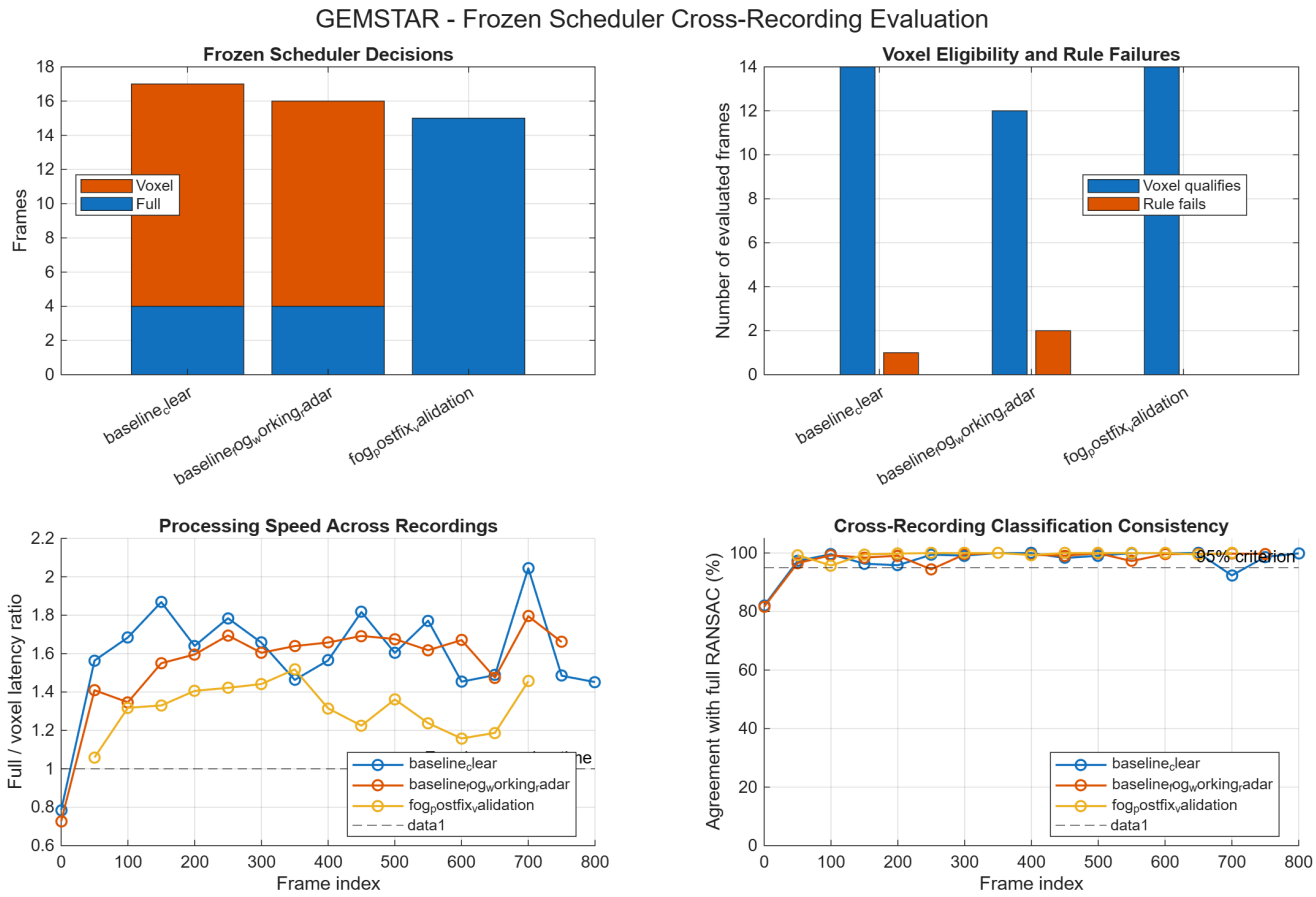
heldout scheduler failure audit



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Research Figure 8

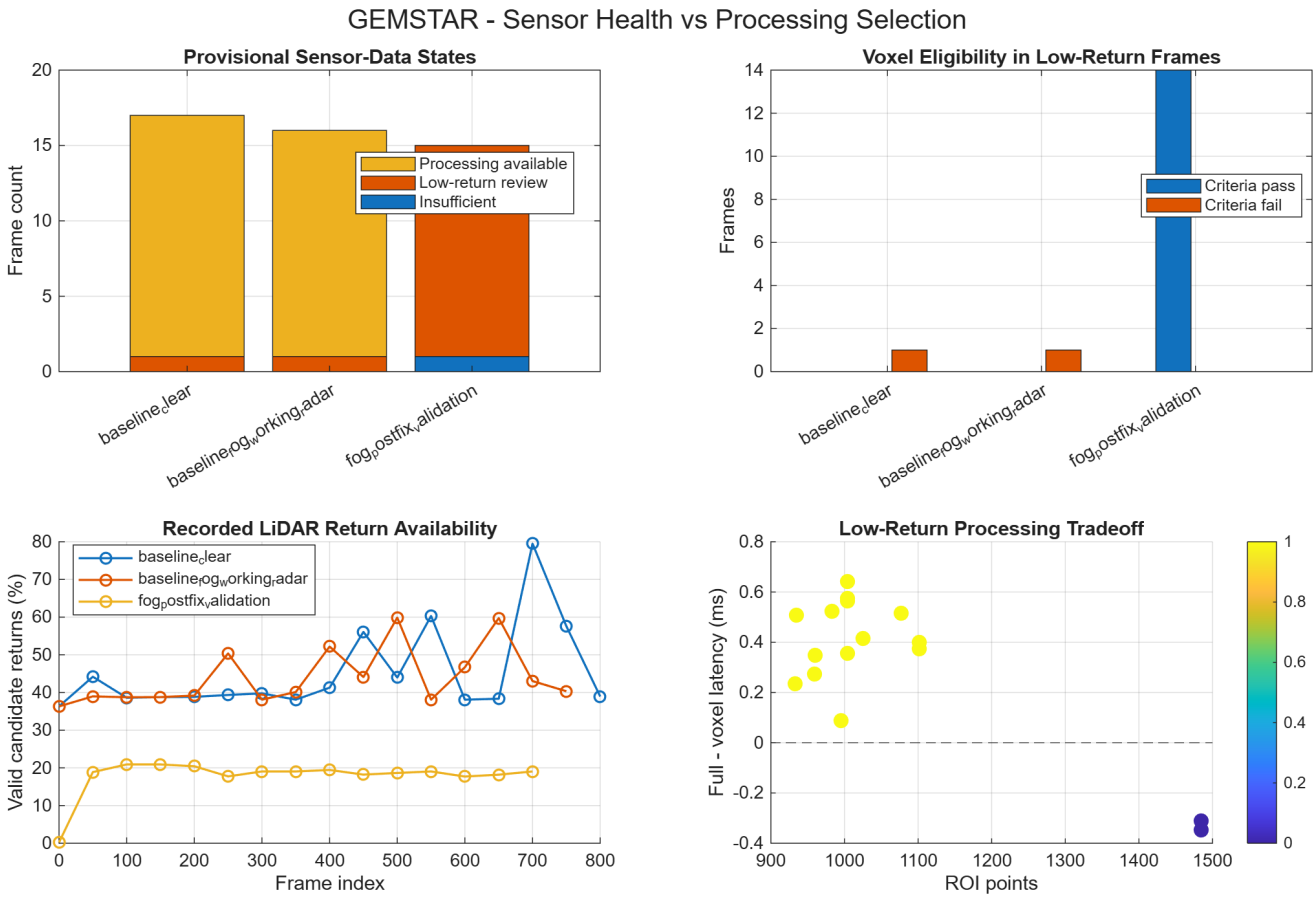
cross recording scheduler evaluation



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Research Figure 9

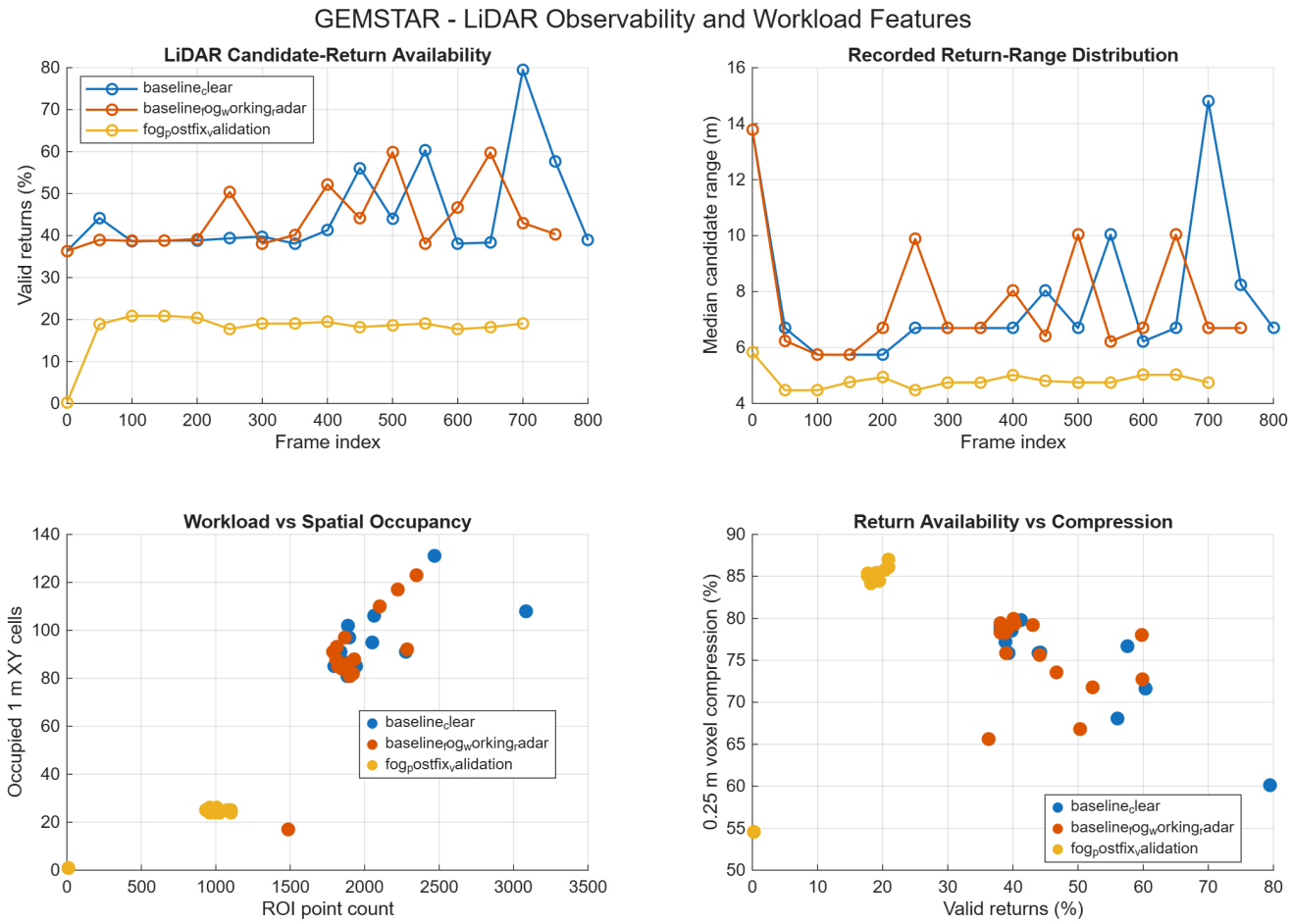
sensor health policy audit



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Research Figure 10

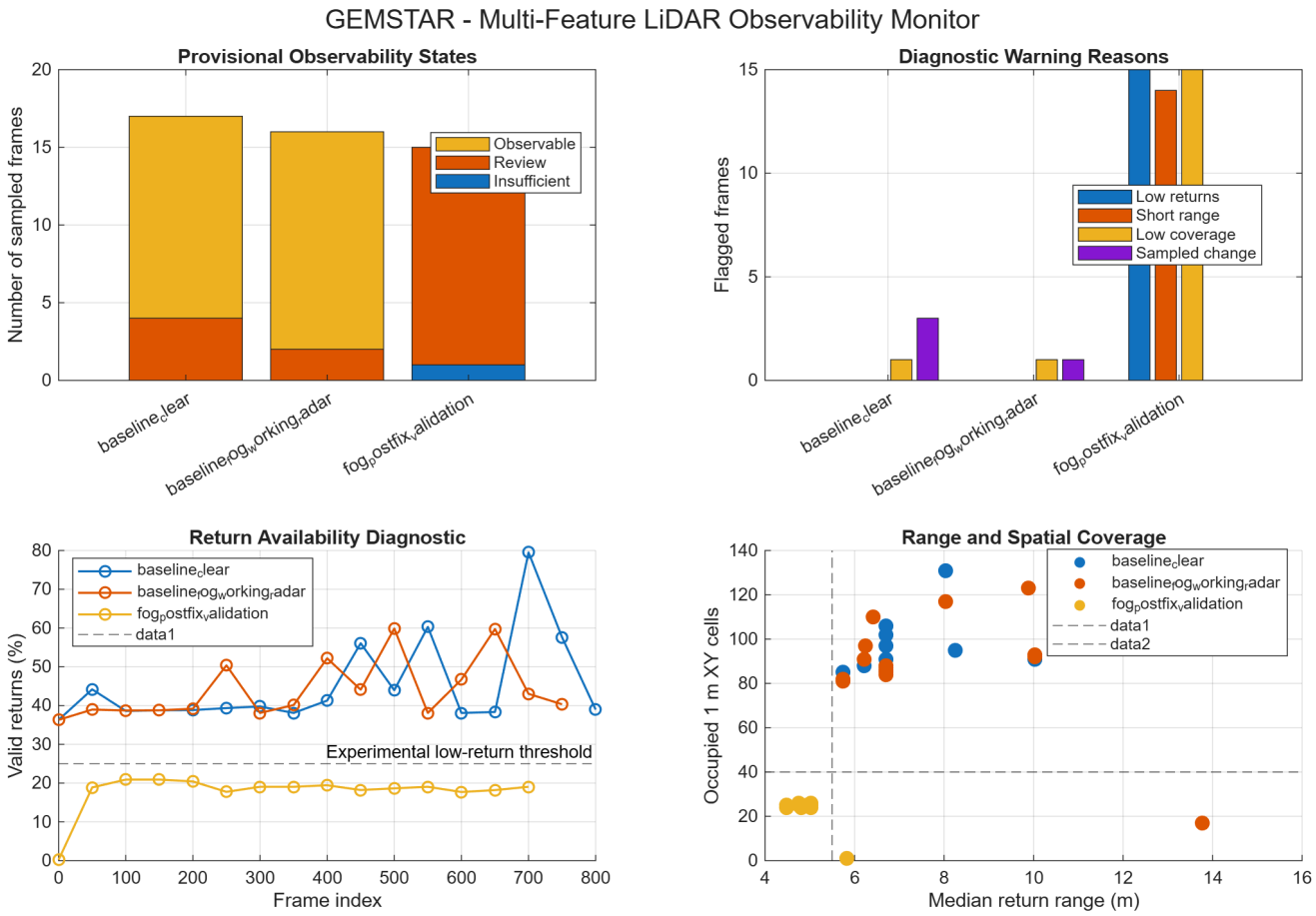
lidar observability features



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Research Figure 11

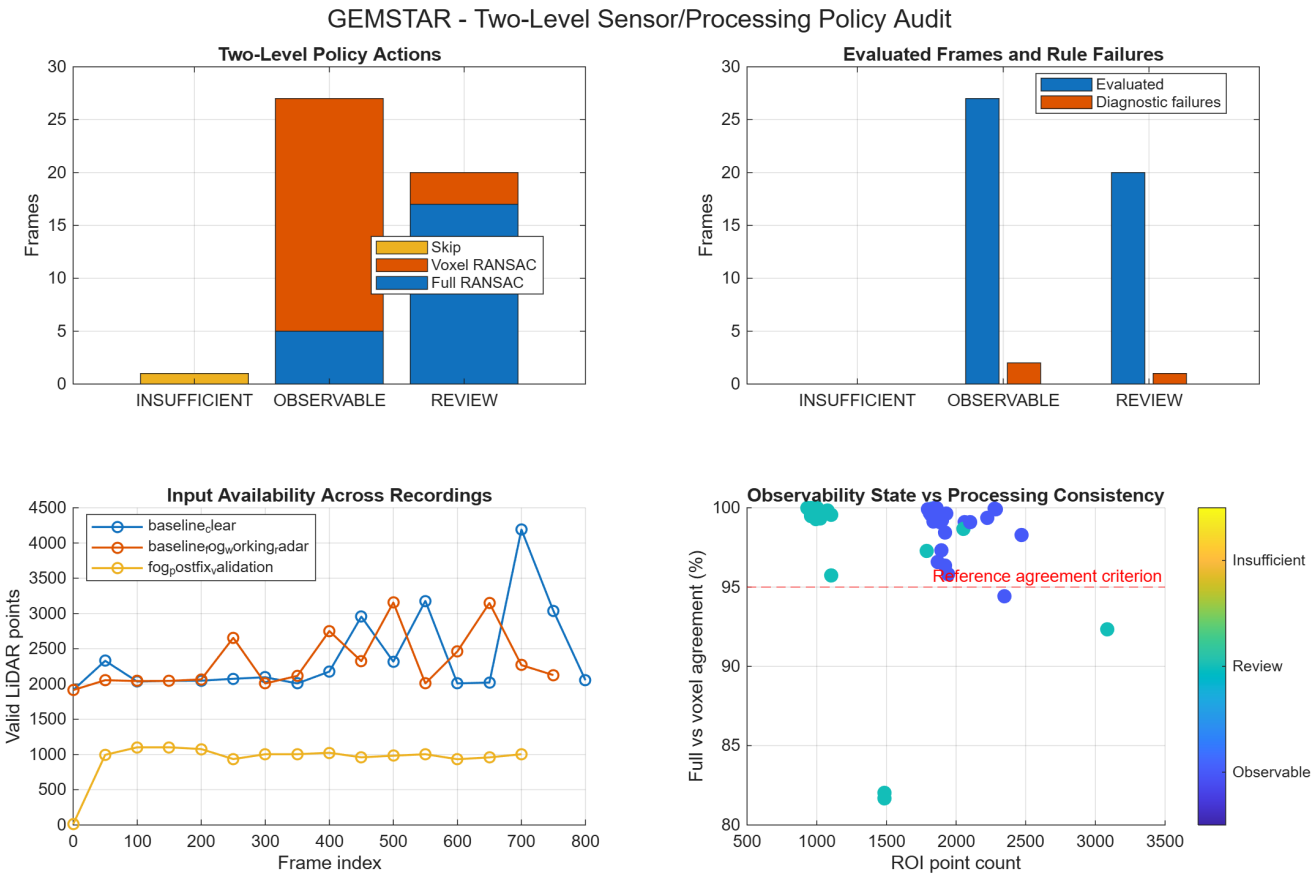
observability monitor



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Research Figure 12

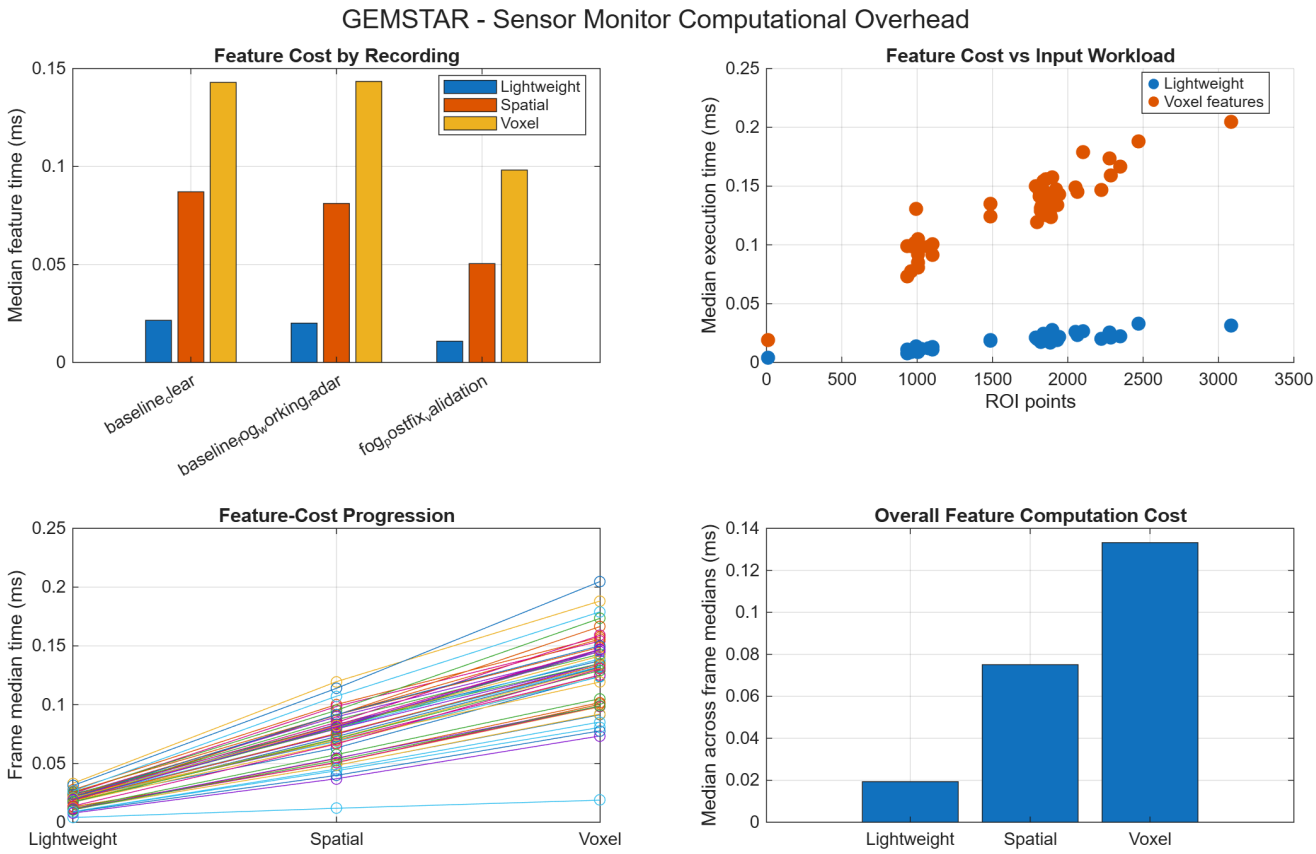
two level policy audit



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Research Figure 13

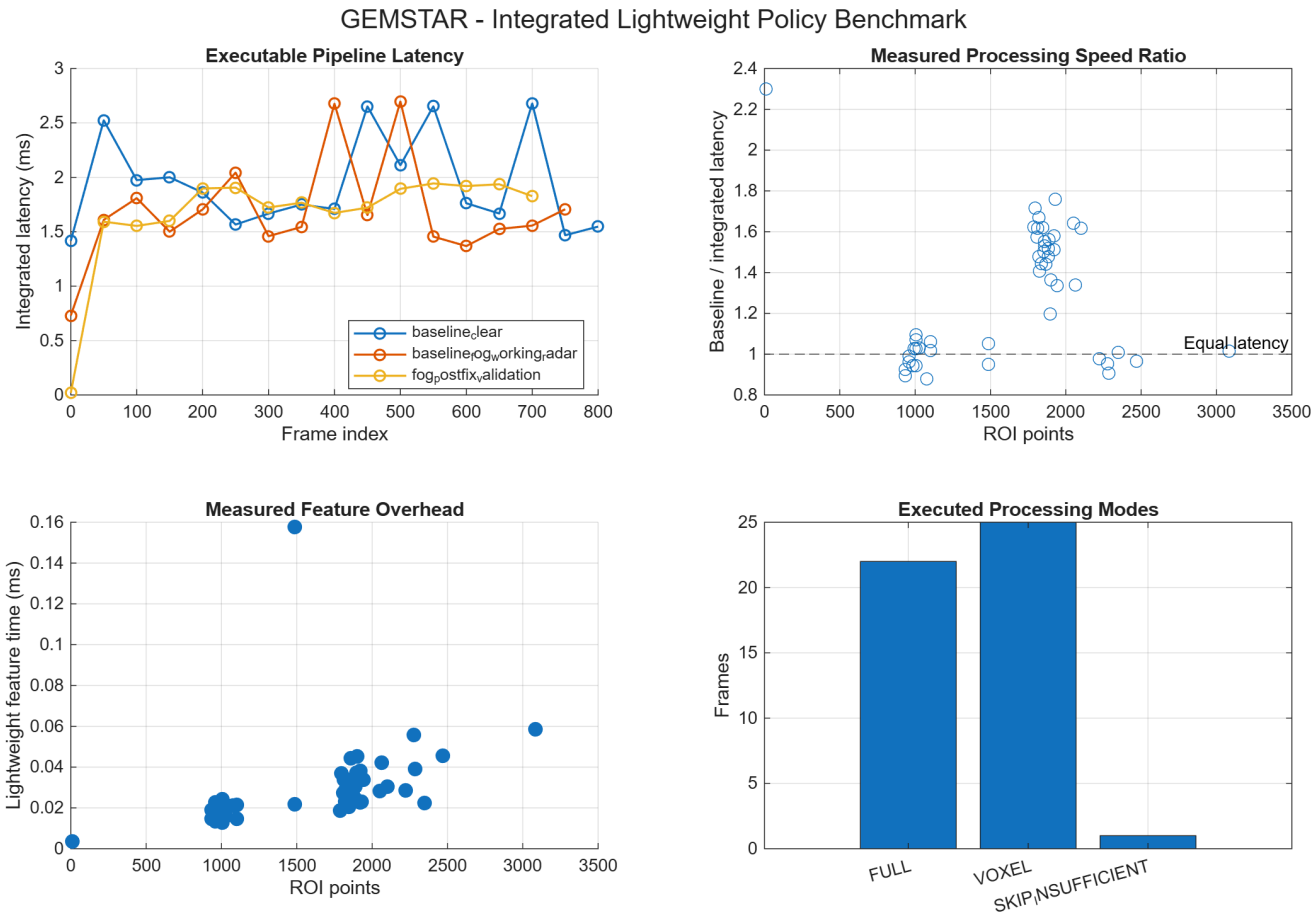
feature overhead benchmark



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Research Figure 14

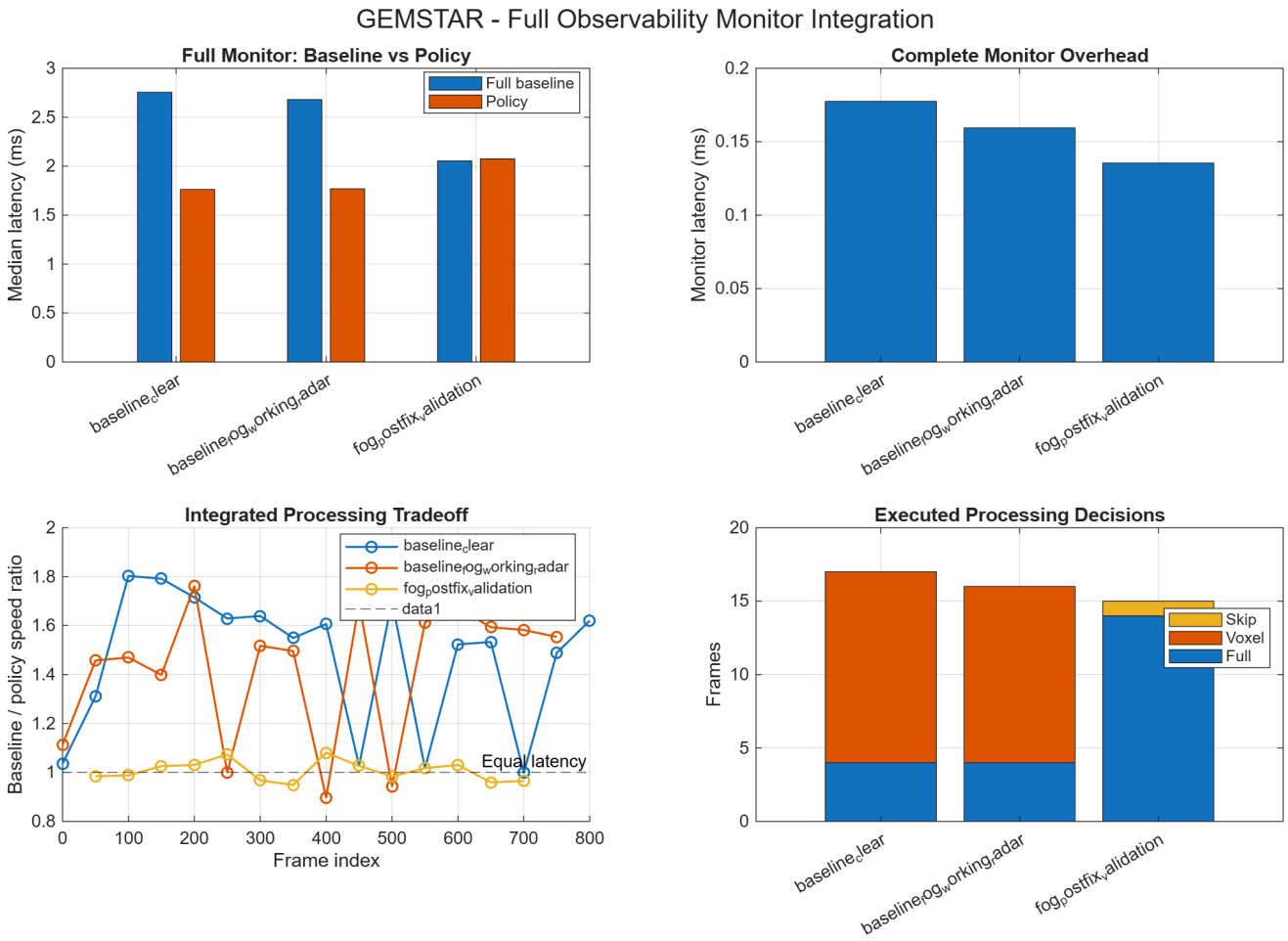
integrated pipeline benchmark



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Research Figure 15

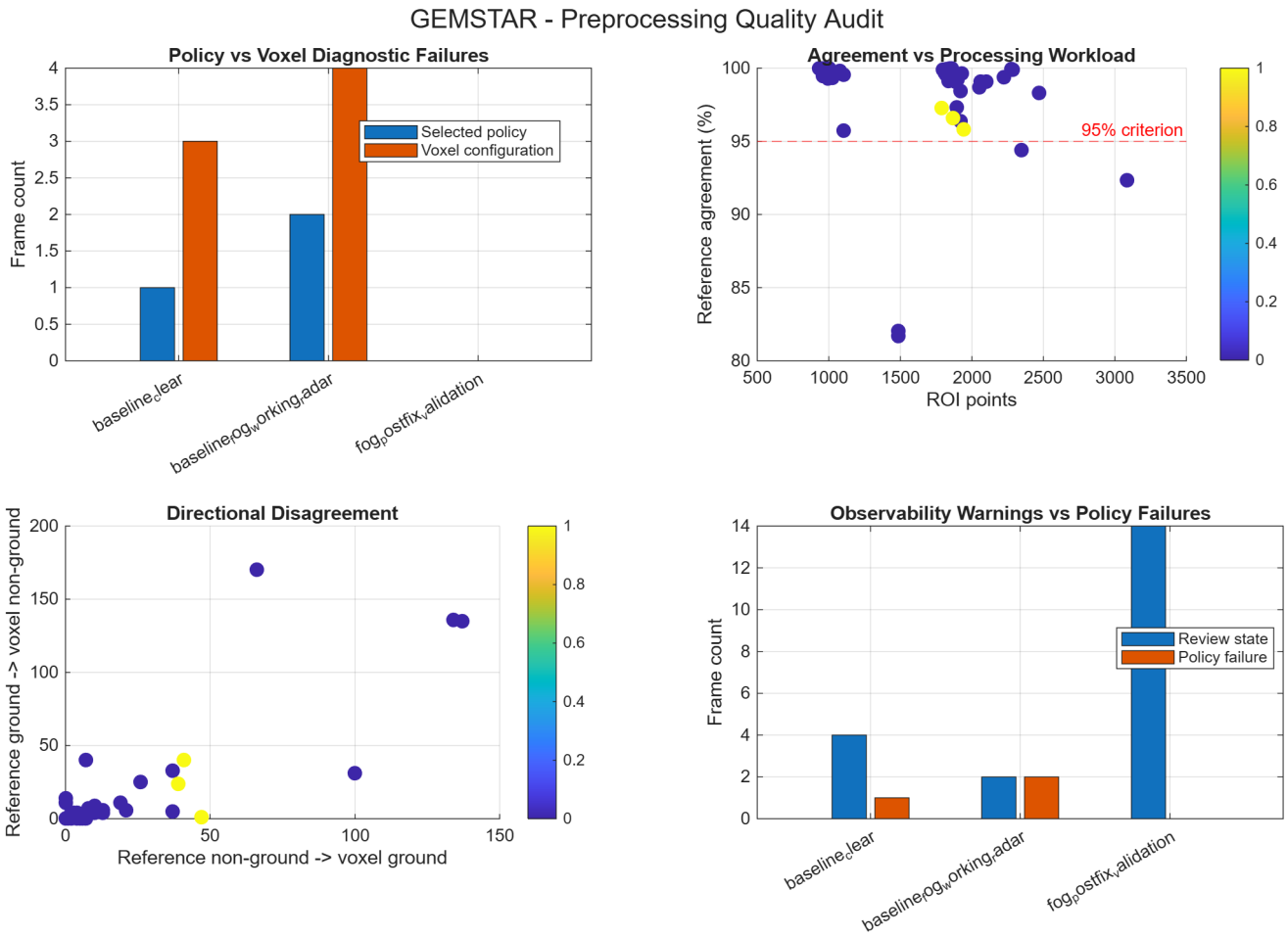
full monitor integration



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Research Figure 16

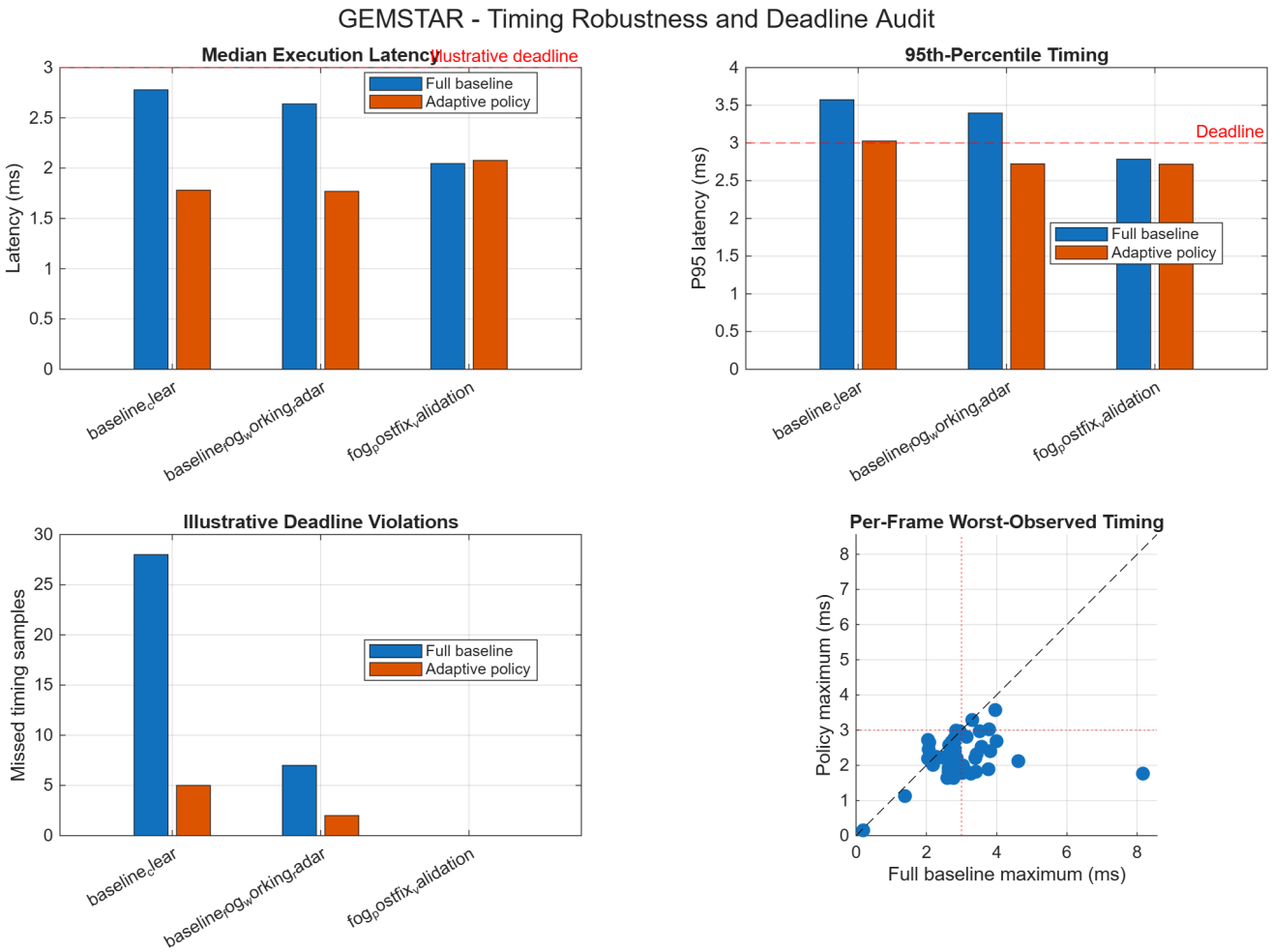
preprocessing quality audit



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Research Figure 17

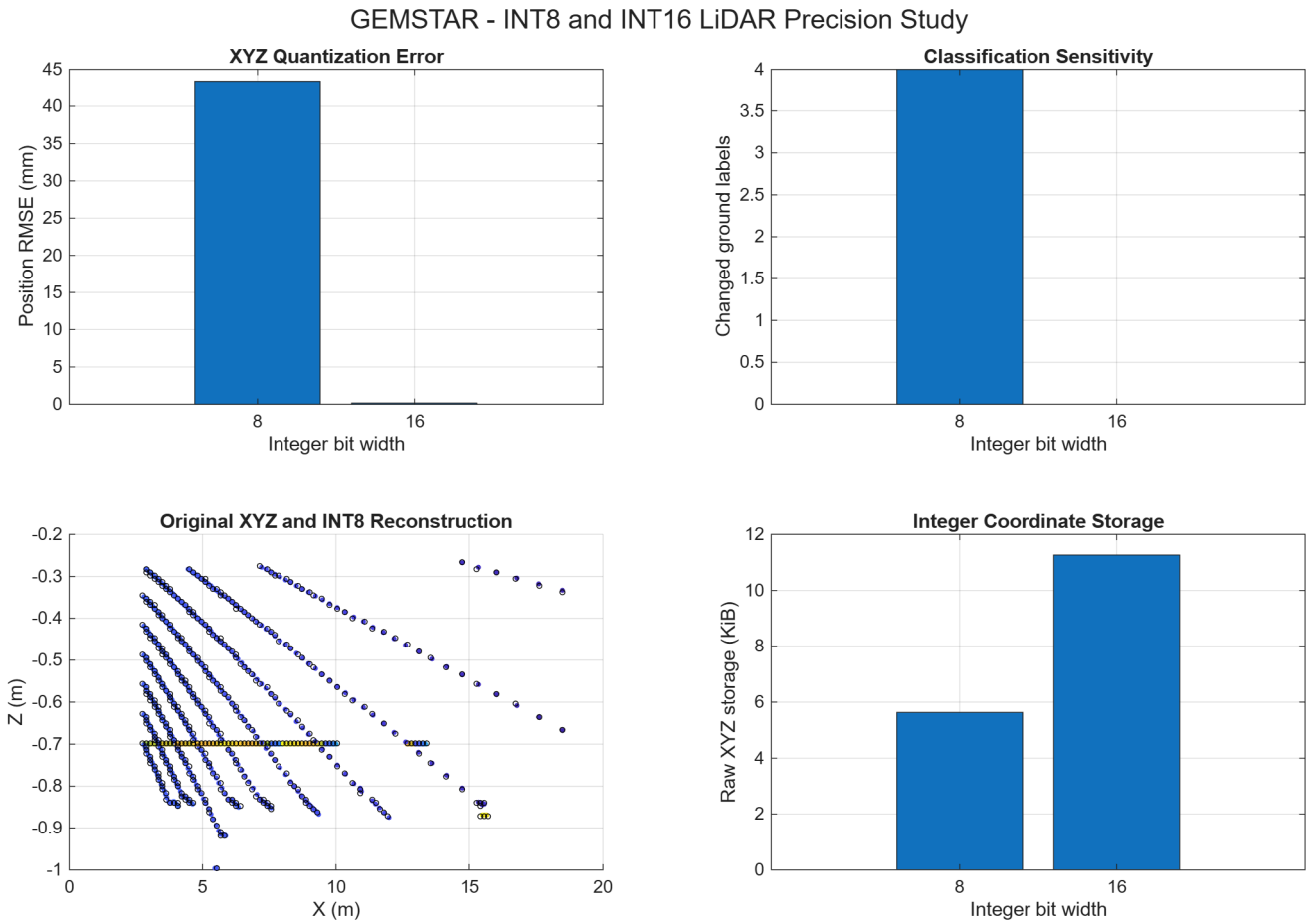
timing deadline analysis



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Research Figure 18

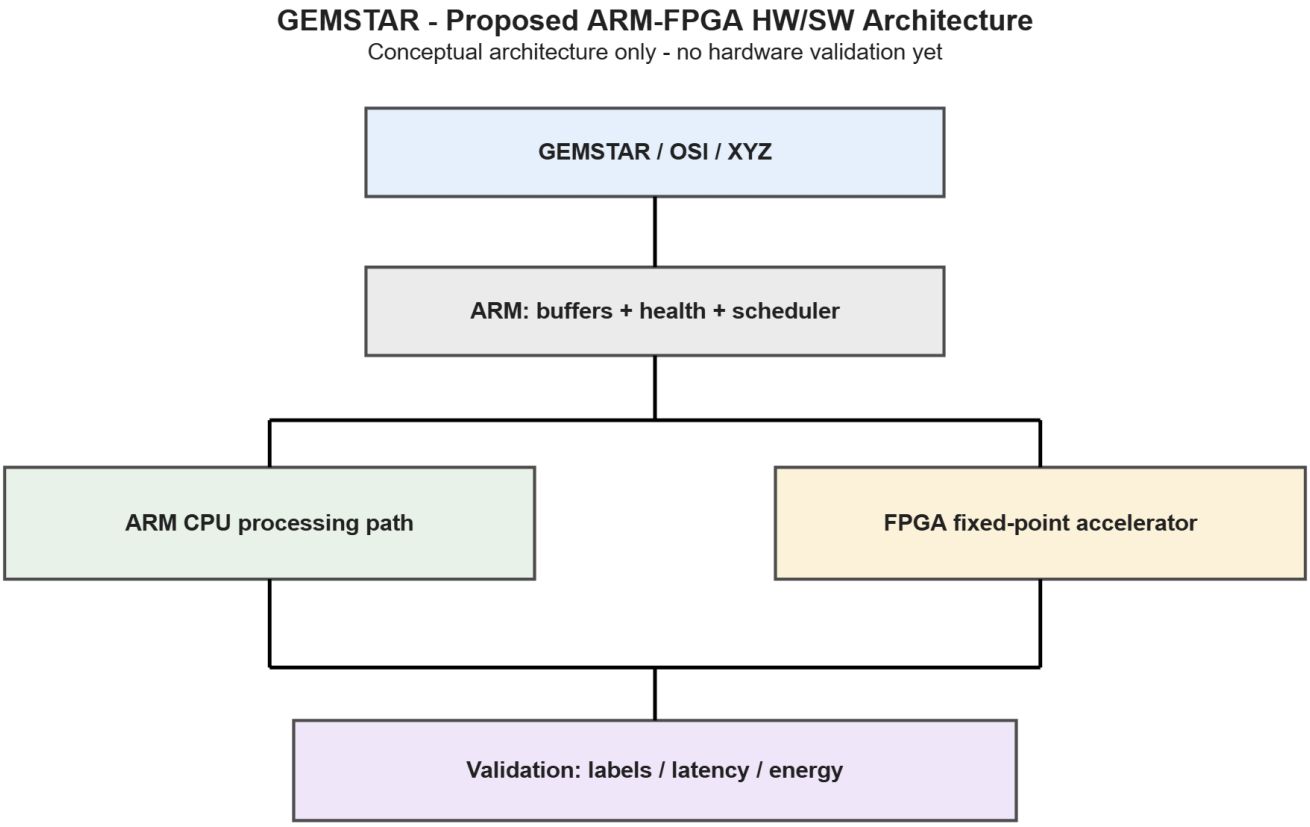
quantization comparison



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Research Figure 19

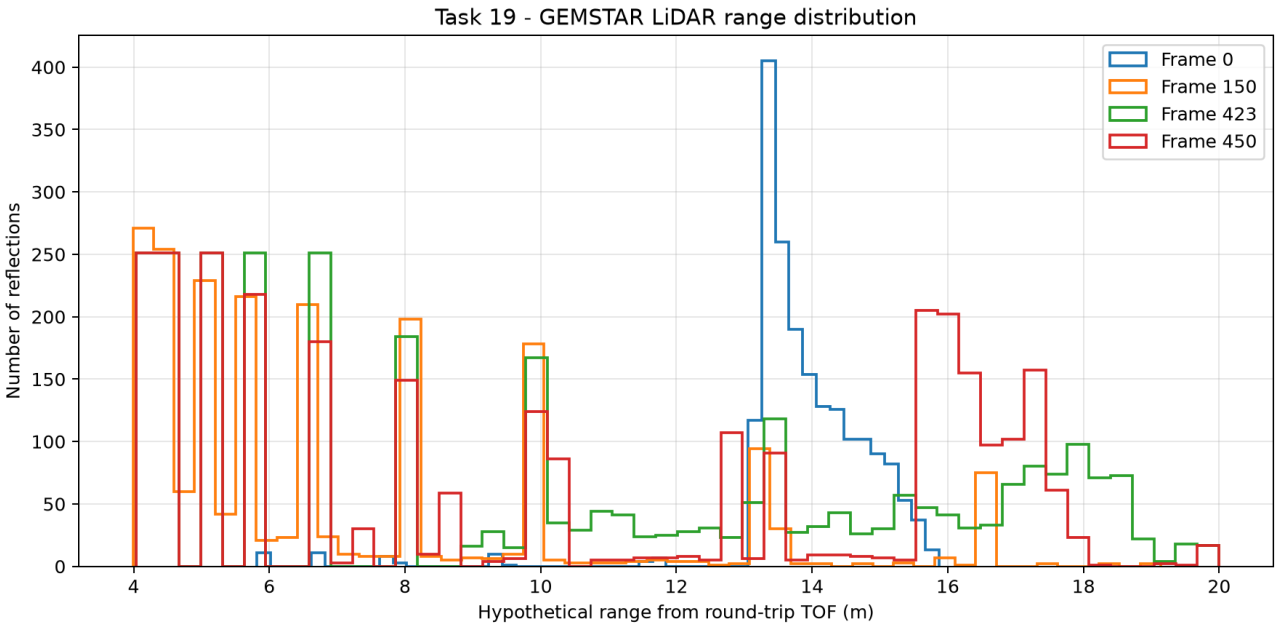
arm fpga architecture



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Research Figure 20

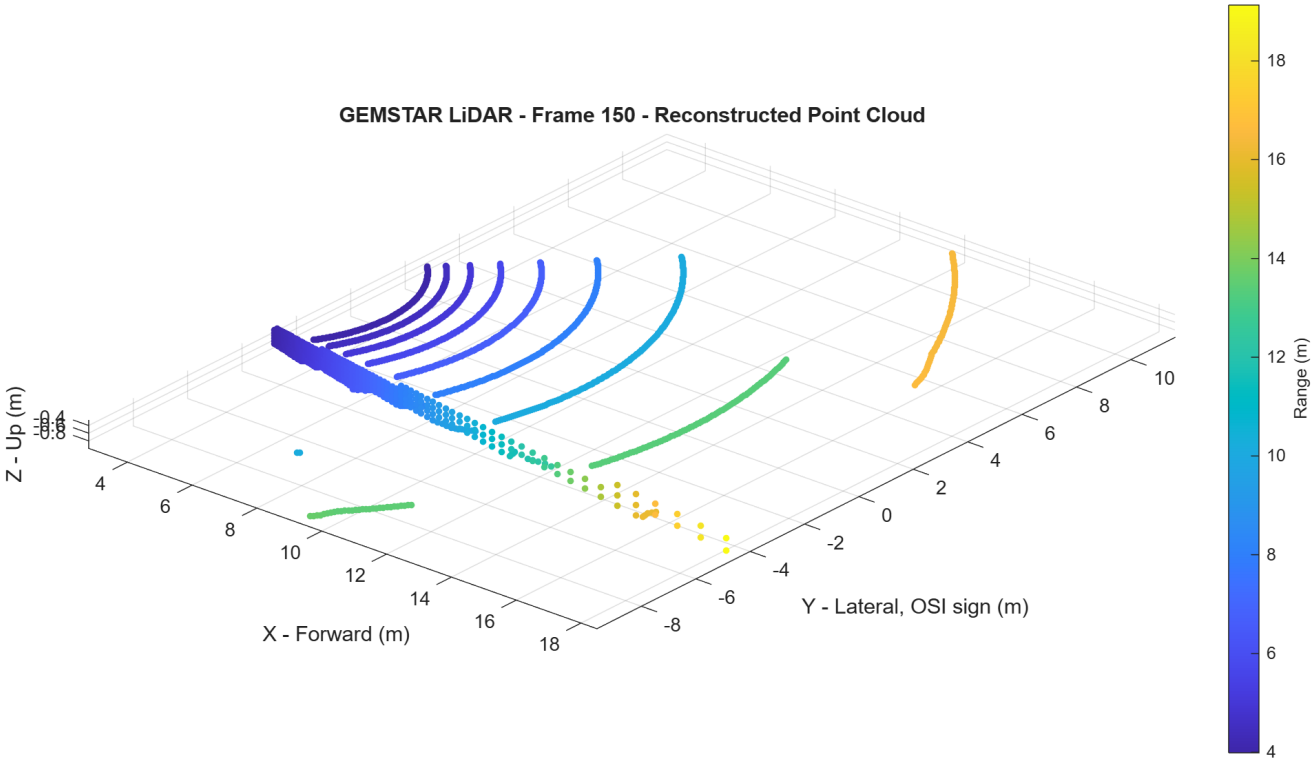
tof range distribution



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Research Figure 21

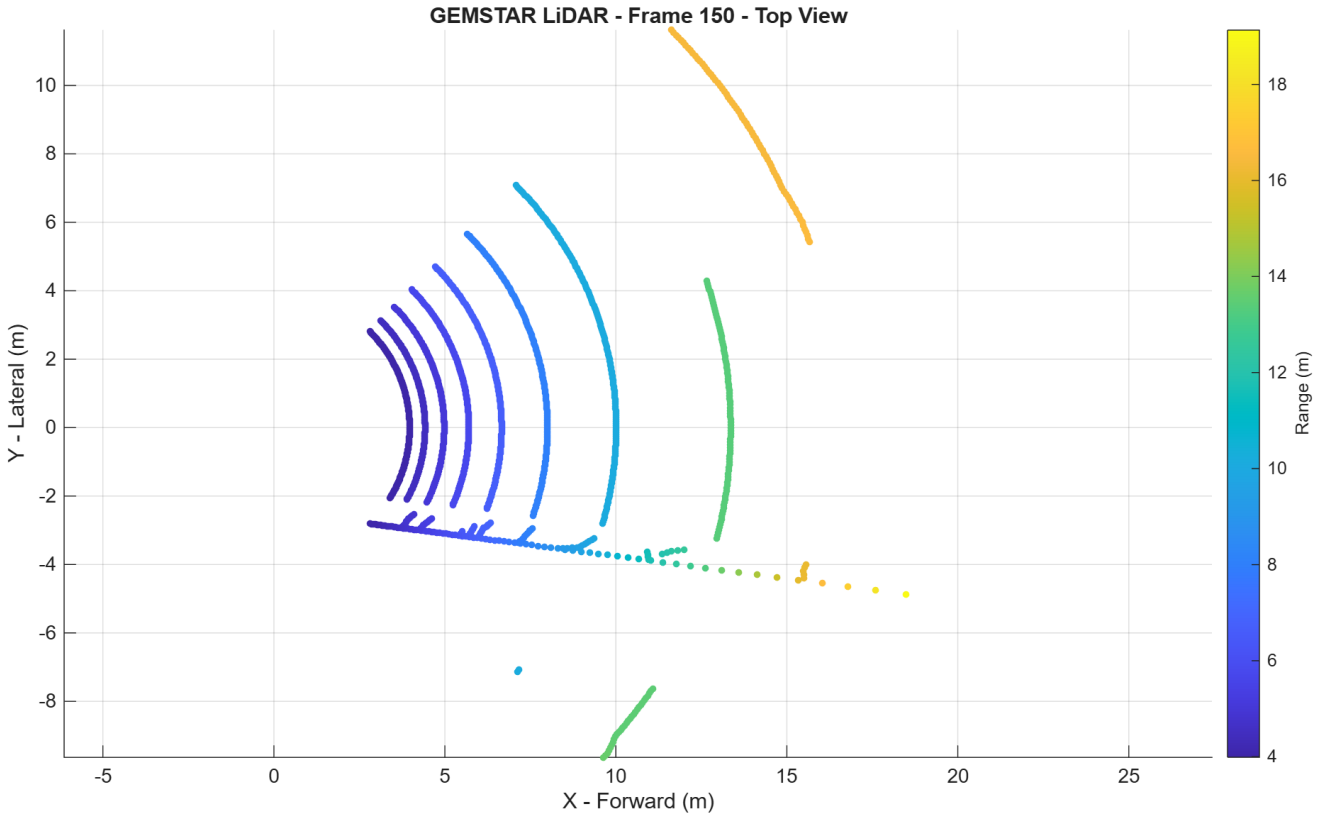
matlab lidar 3d



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Research Figure 22

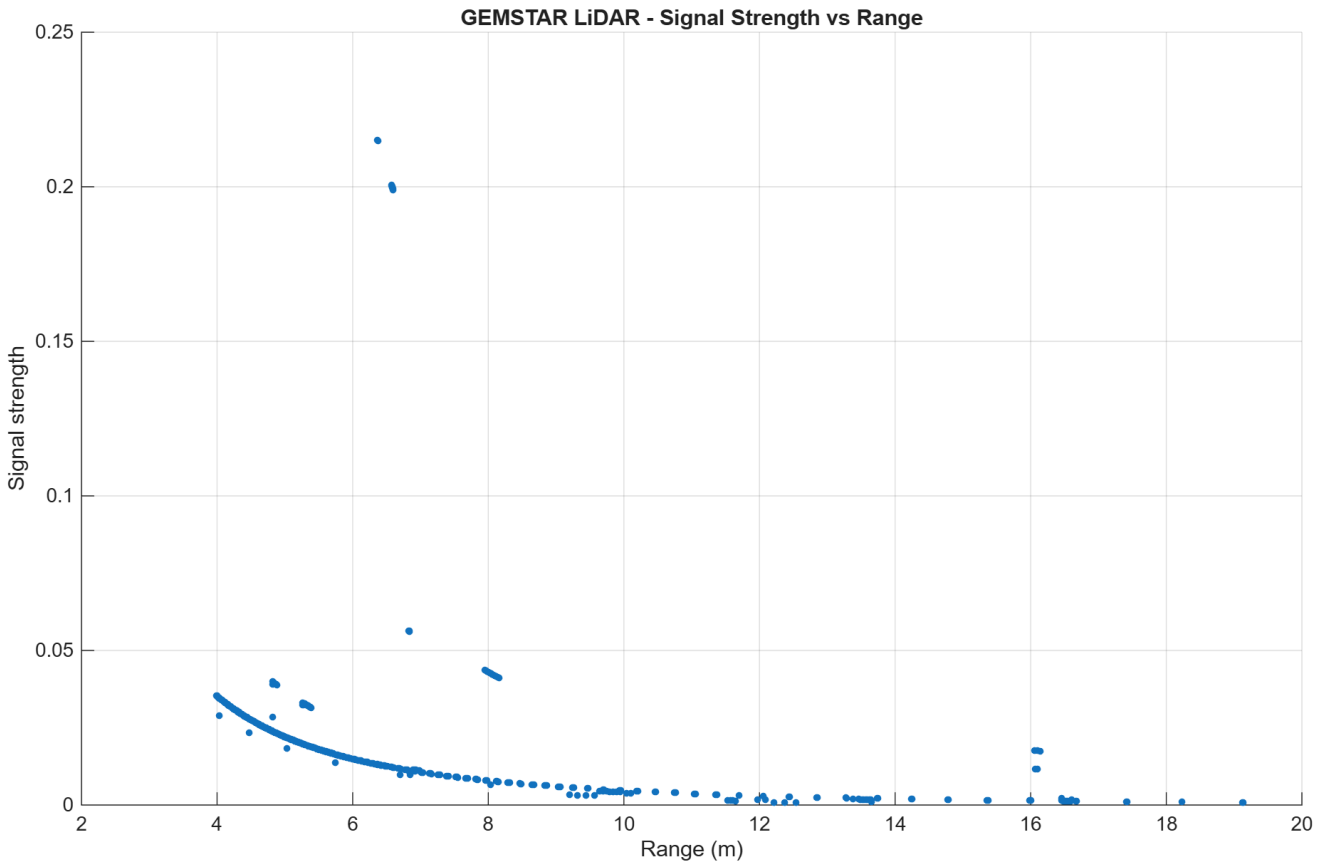
matlab lidar top view



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Research Figure 23

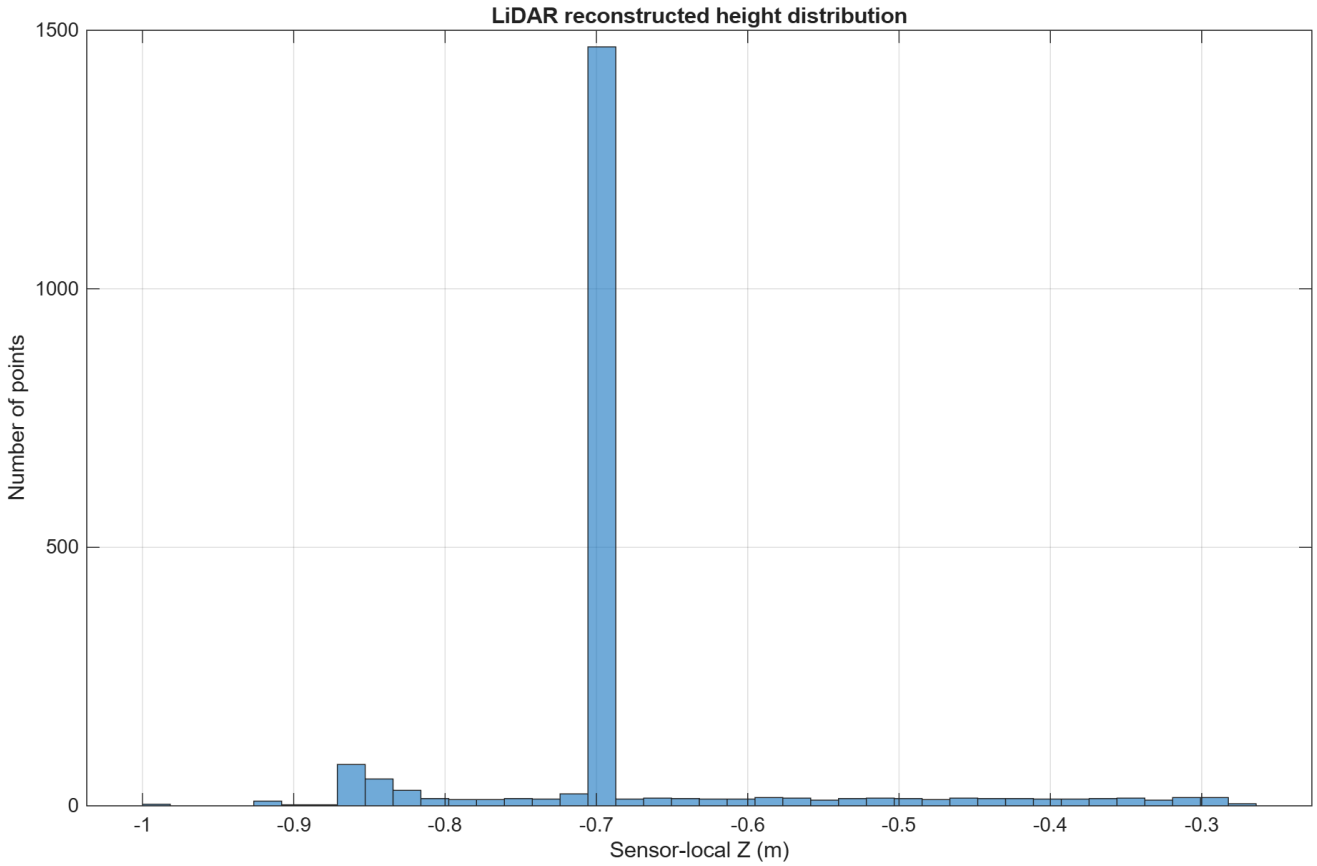
matlab signal strength



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Research Figure 24

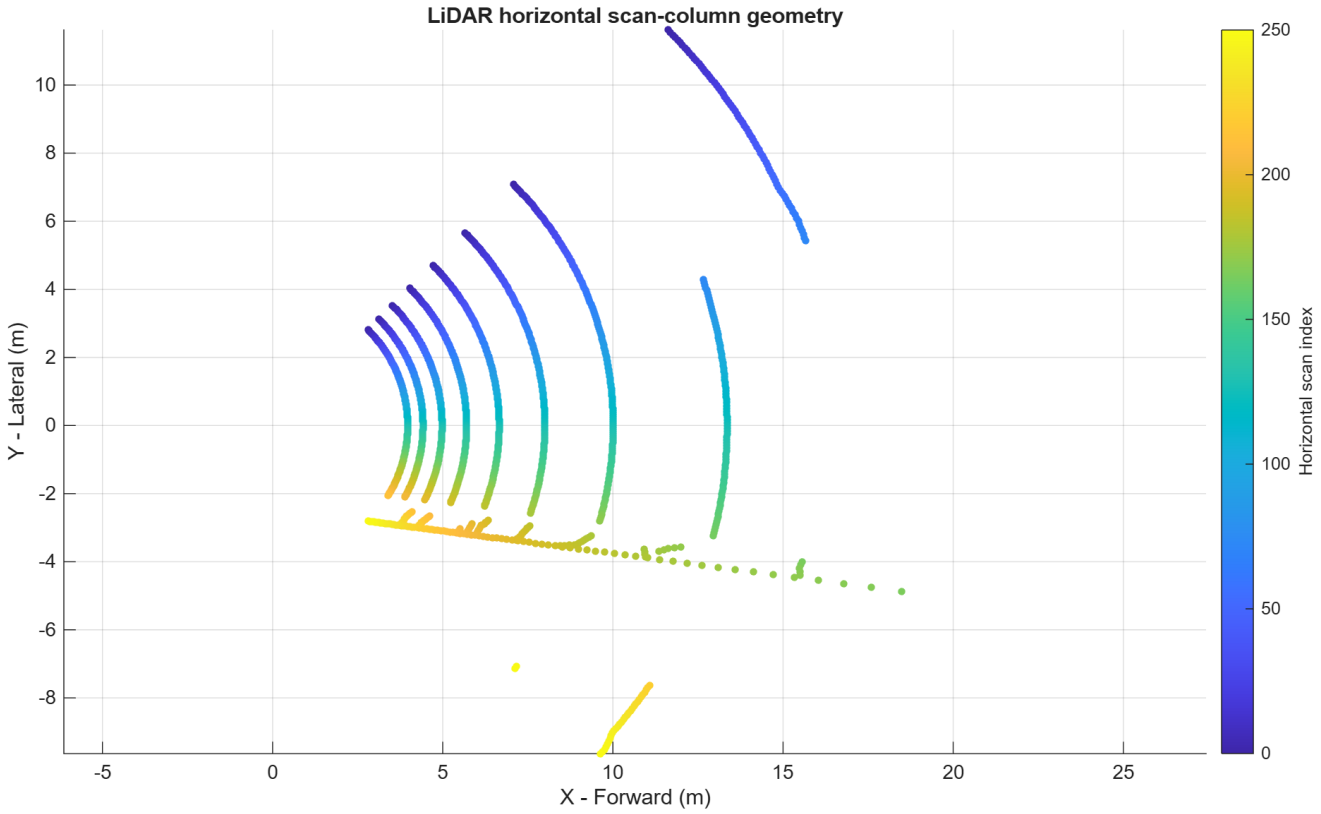
height histogram



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Research Figure 25

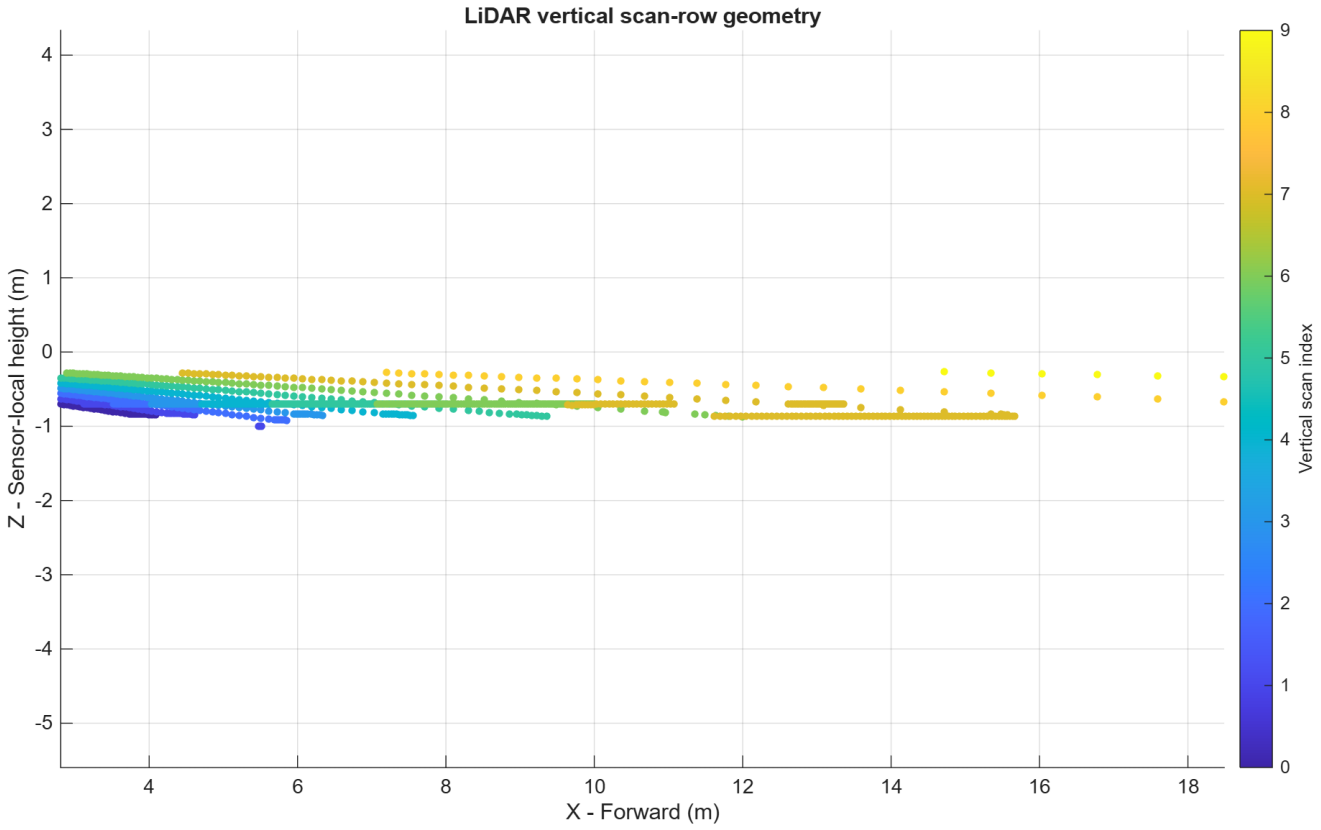
horizontal geometry



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Research Figure 26

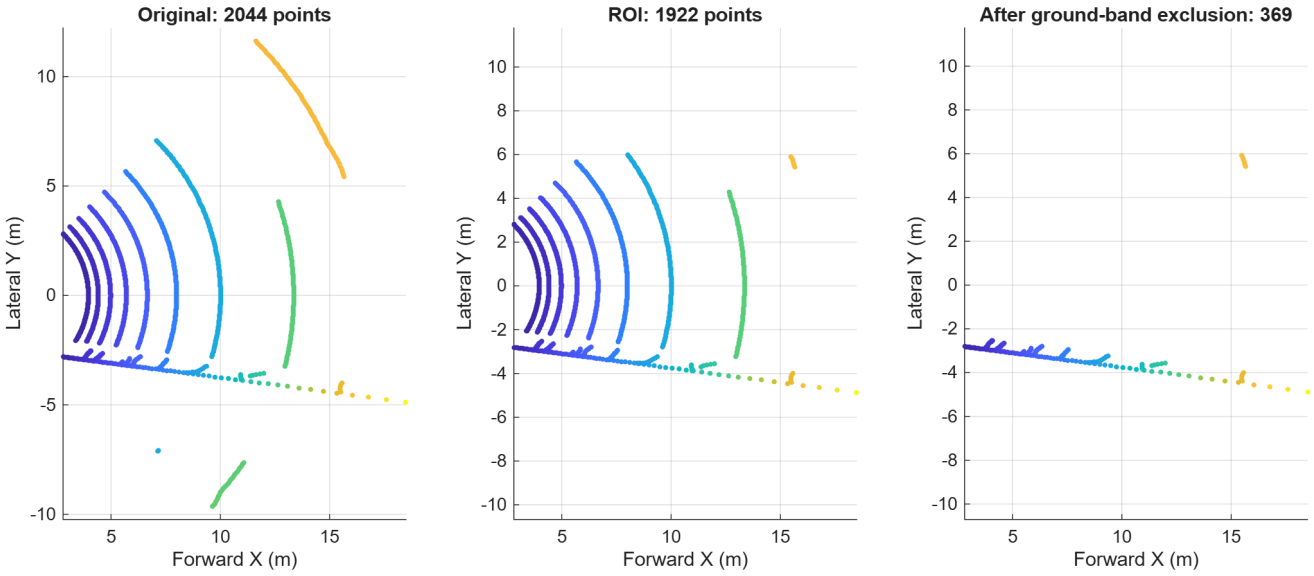
vertical geometry



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Research Figure 27

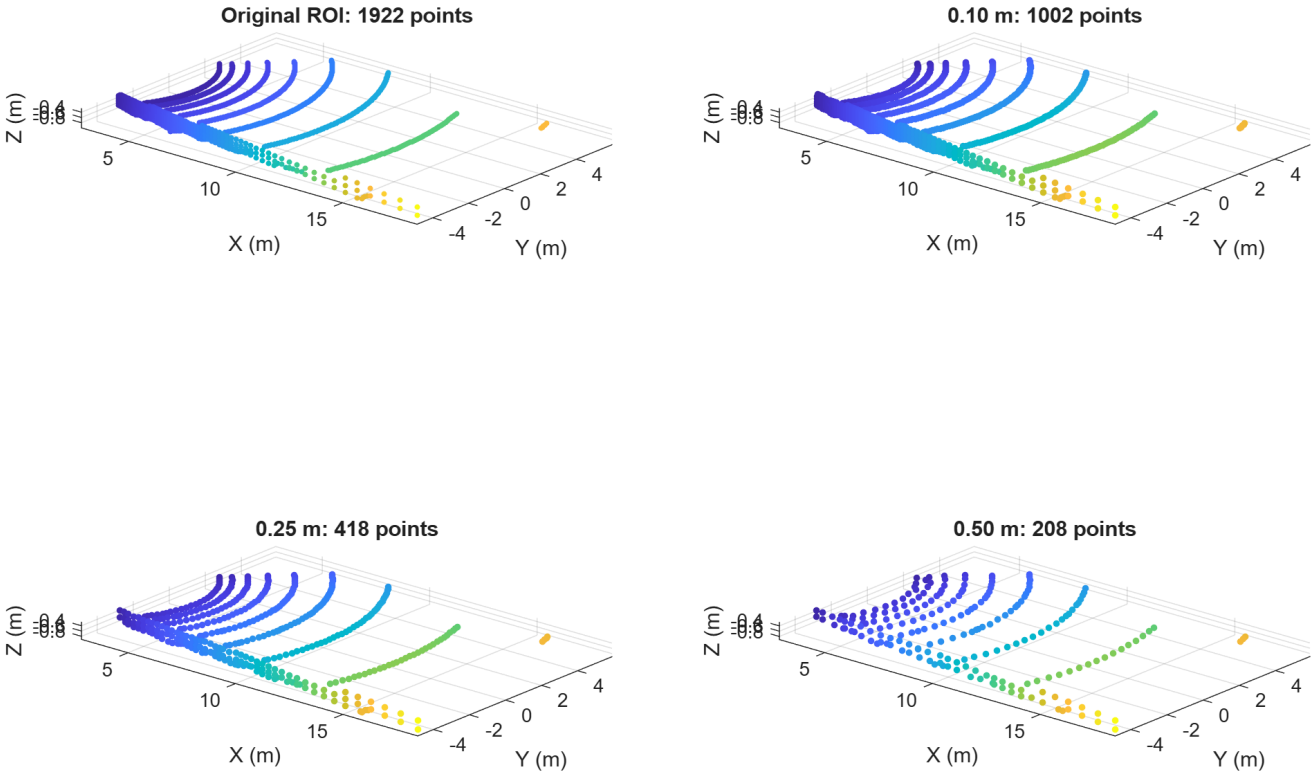
roi ground comparison



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Research Figure 28

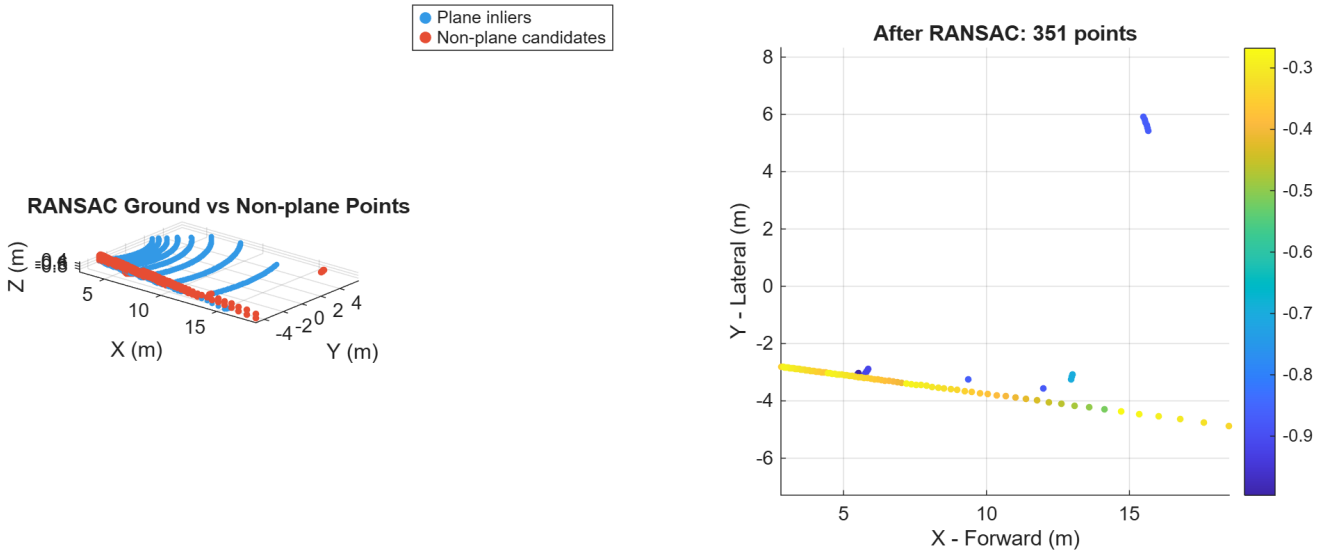
voxel comparison



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Research Figure 29

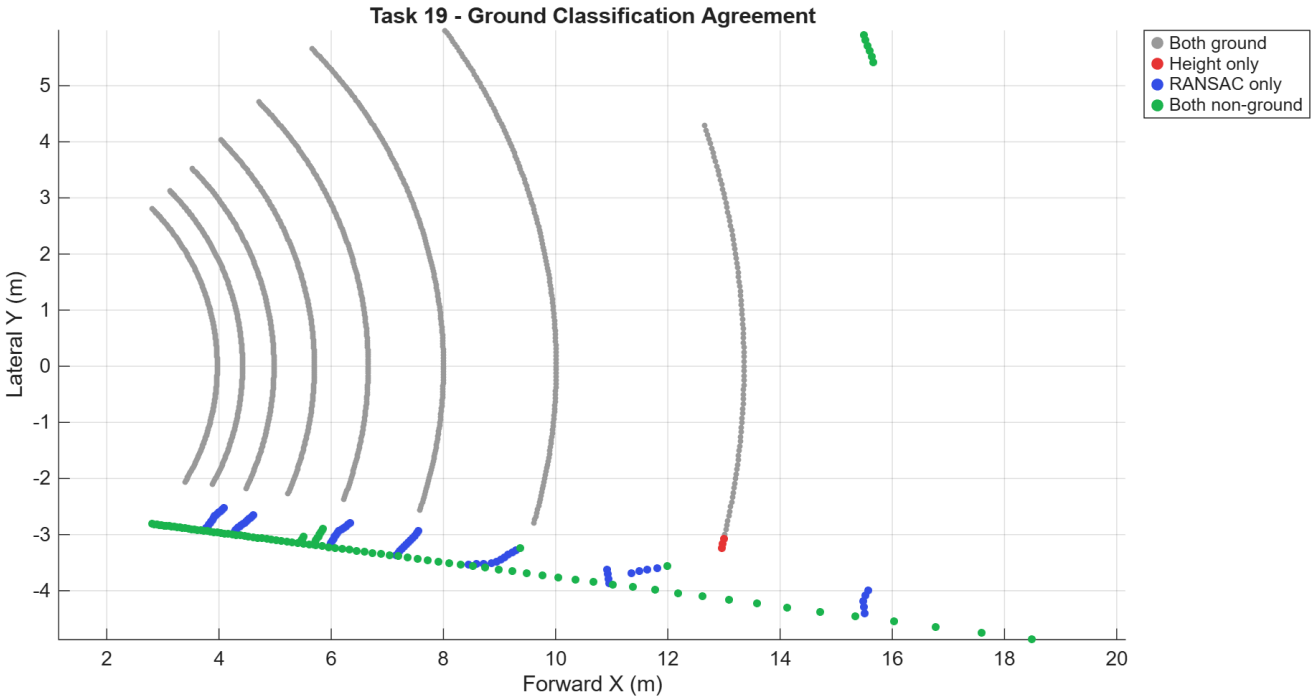
ransac ground segmentation



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Research Figure 30

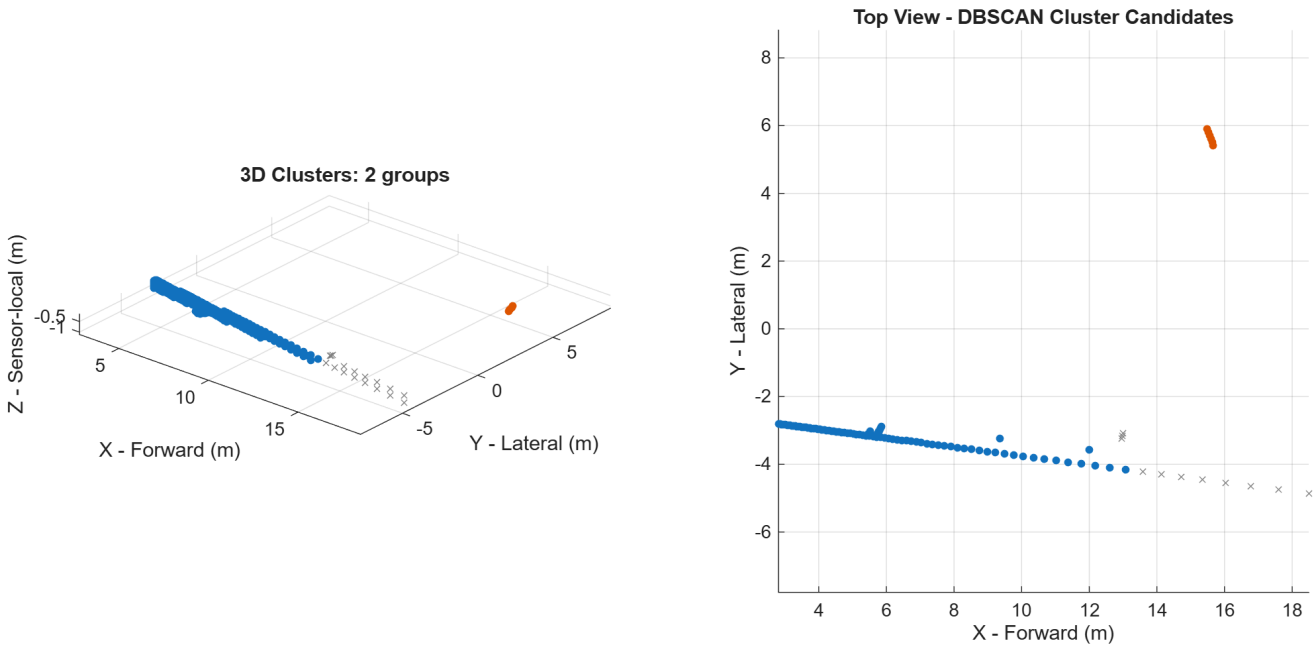
ground method comparison



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Research Figure 31

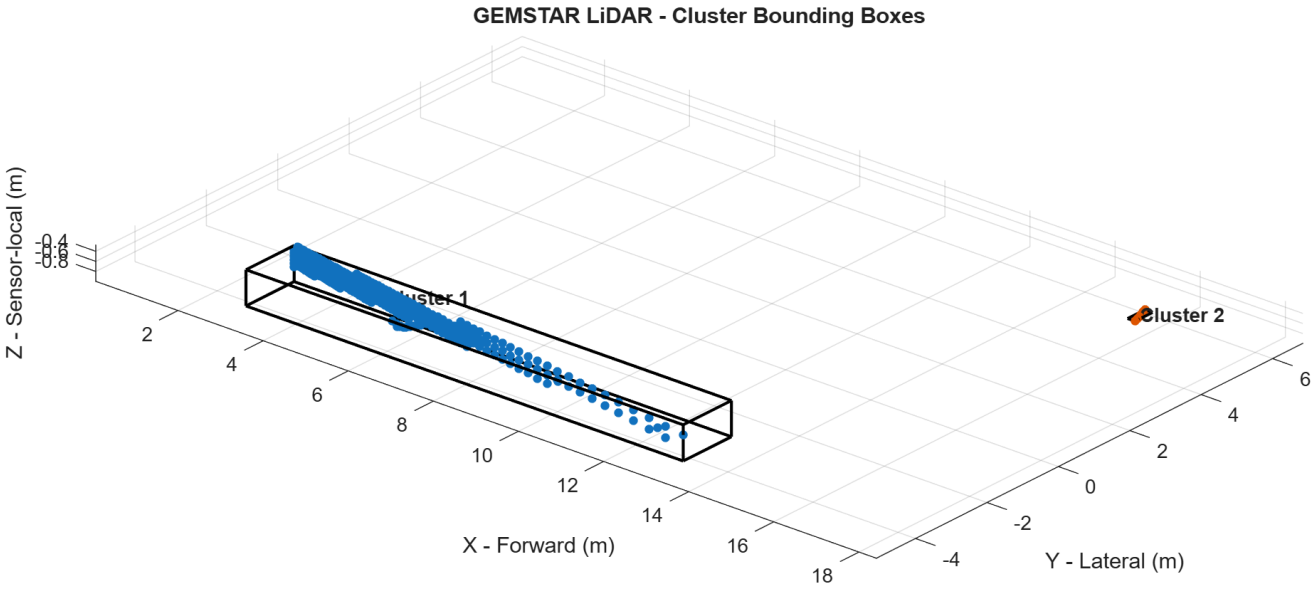
dbscan clustering



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Research Figure 32

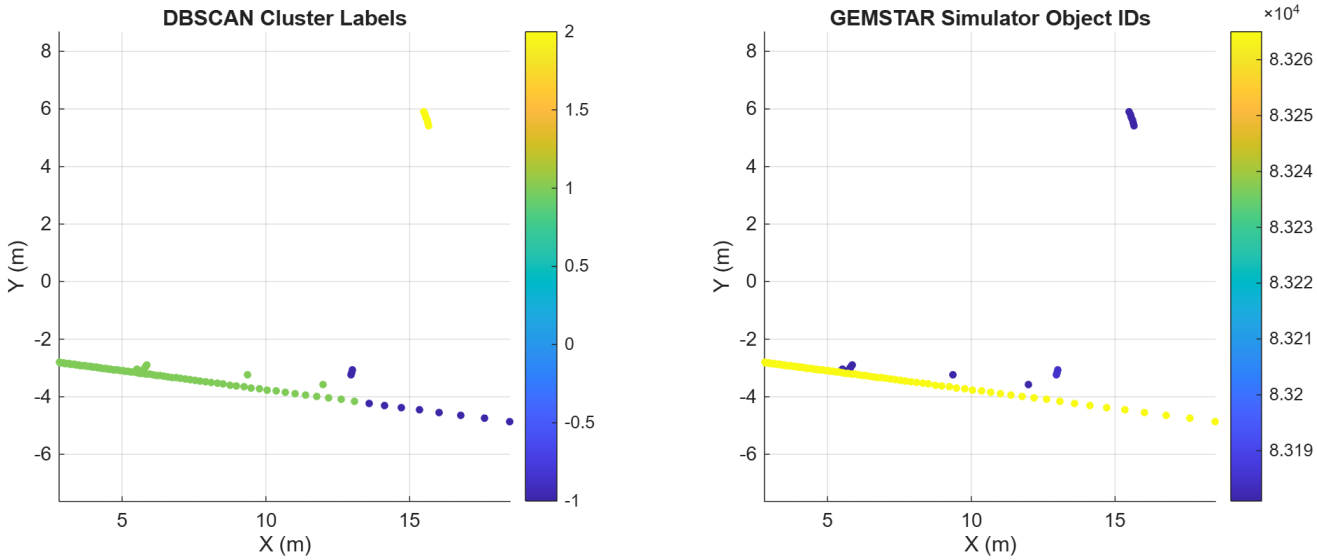
cluster bounding boxes



Original saved graph: 01_Graphs\task19_7b_cluster_bounding_boxes.png

Research Figure 33

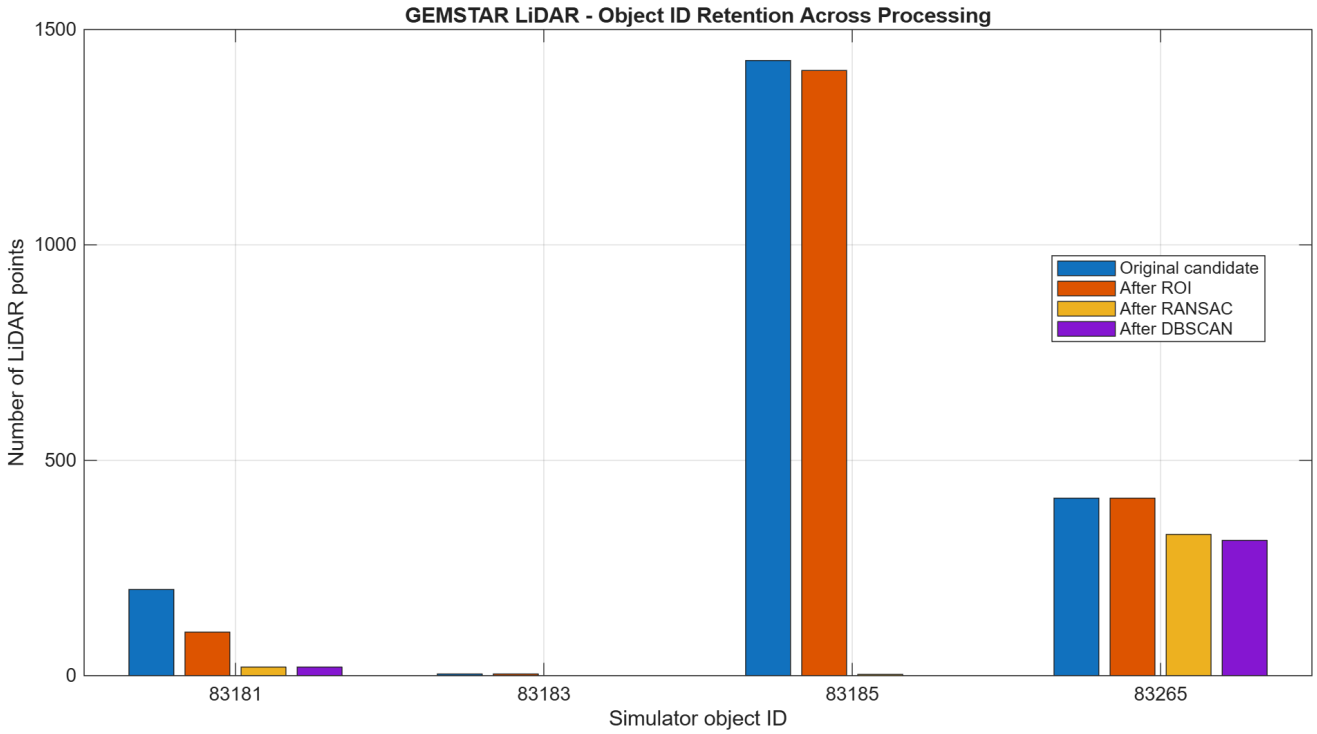
cluster vs object id



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Research Figure 34

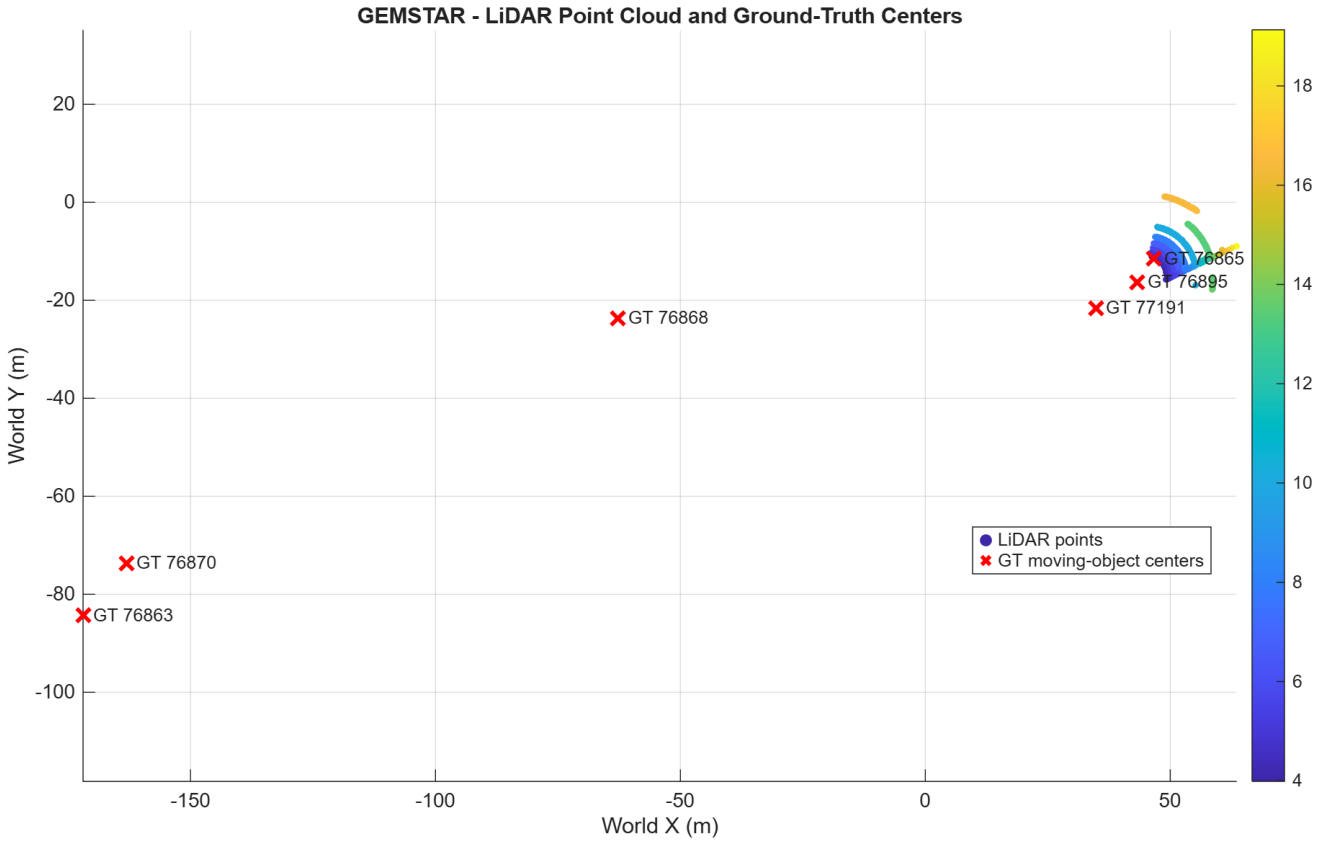
object id retention



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Research Figure 35

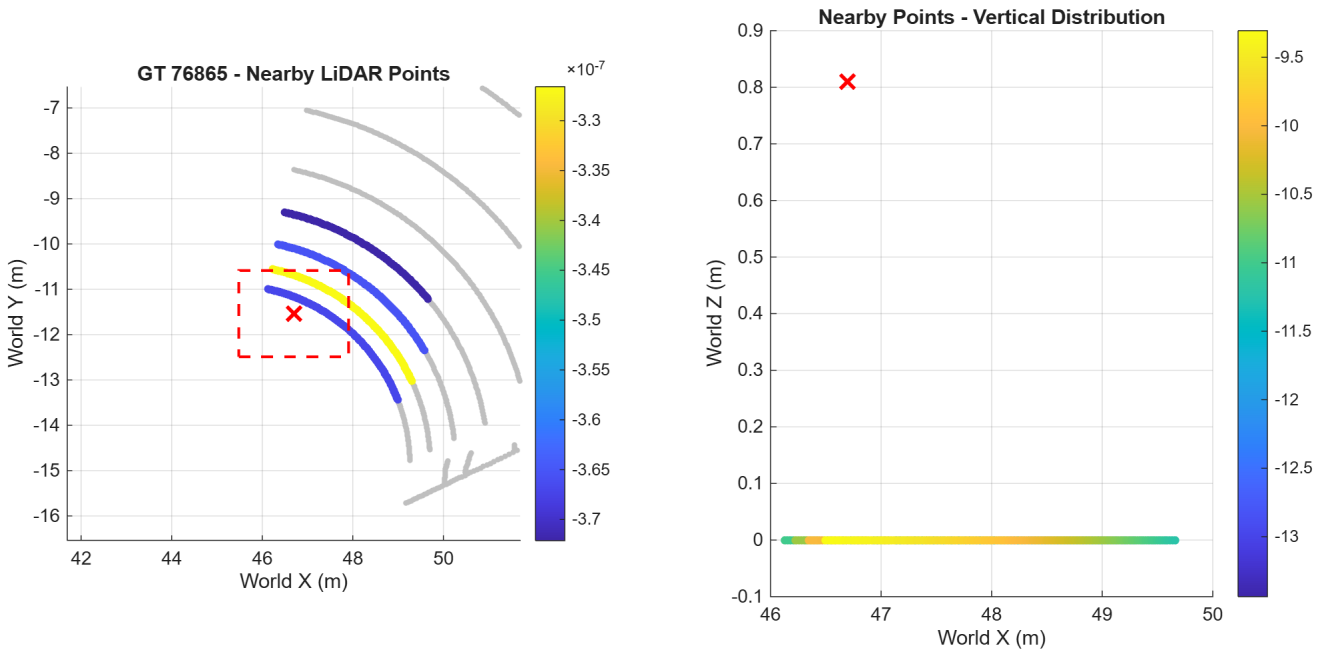
world alignment



Original saved graph: 01_Graphs\task19_8h_world_alignment.png

Research Figure 36

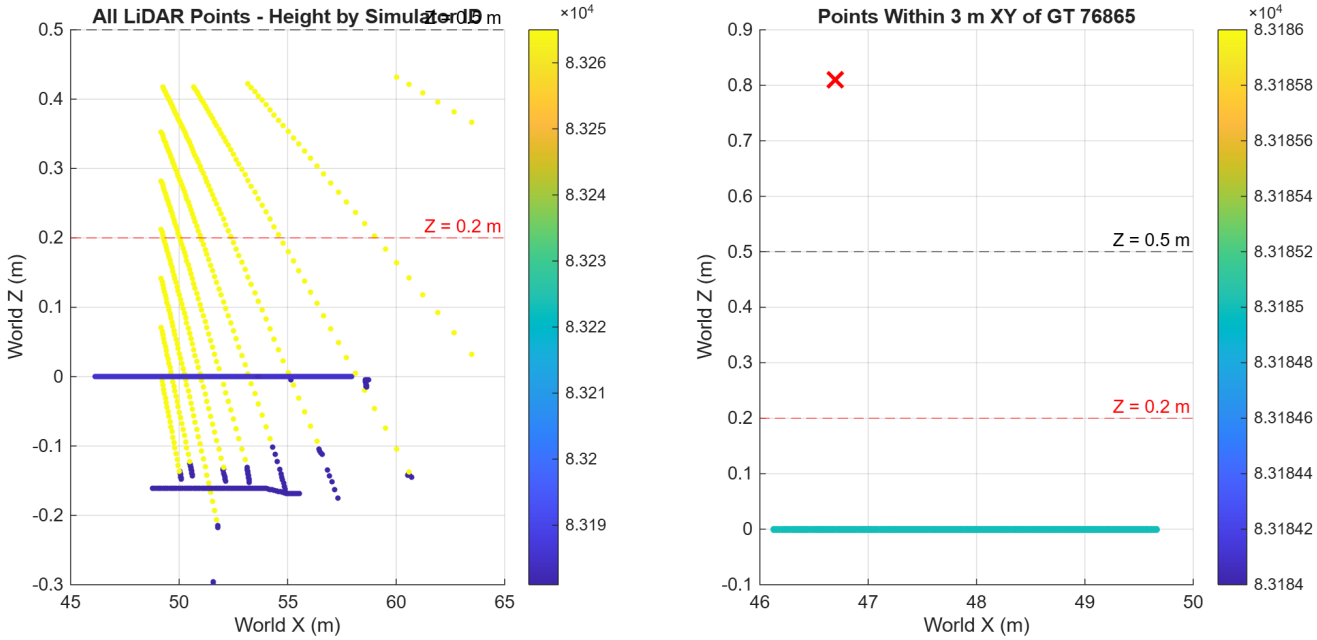
gt76865 proximity analysis



Original saved graph: 01_Graphs\task19_8i_gt76865_proximity_analysis.png

Research Figure 37

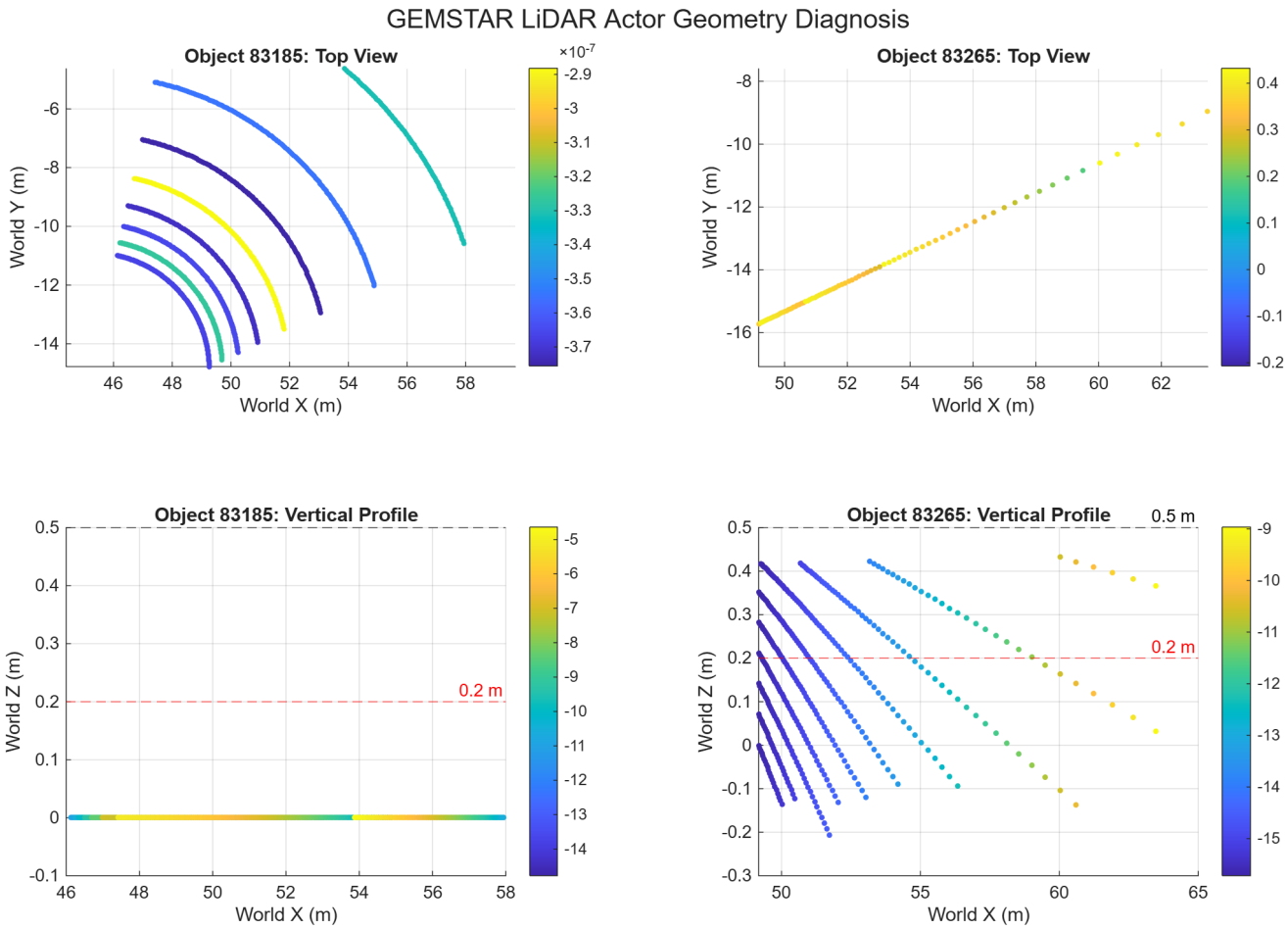
ground height validation



Original saved graph: 01_Graphs\task19_8j_ground_height_validation.png

Research Figure 38

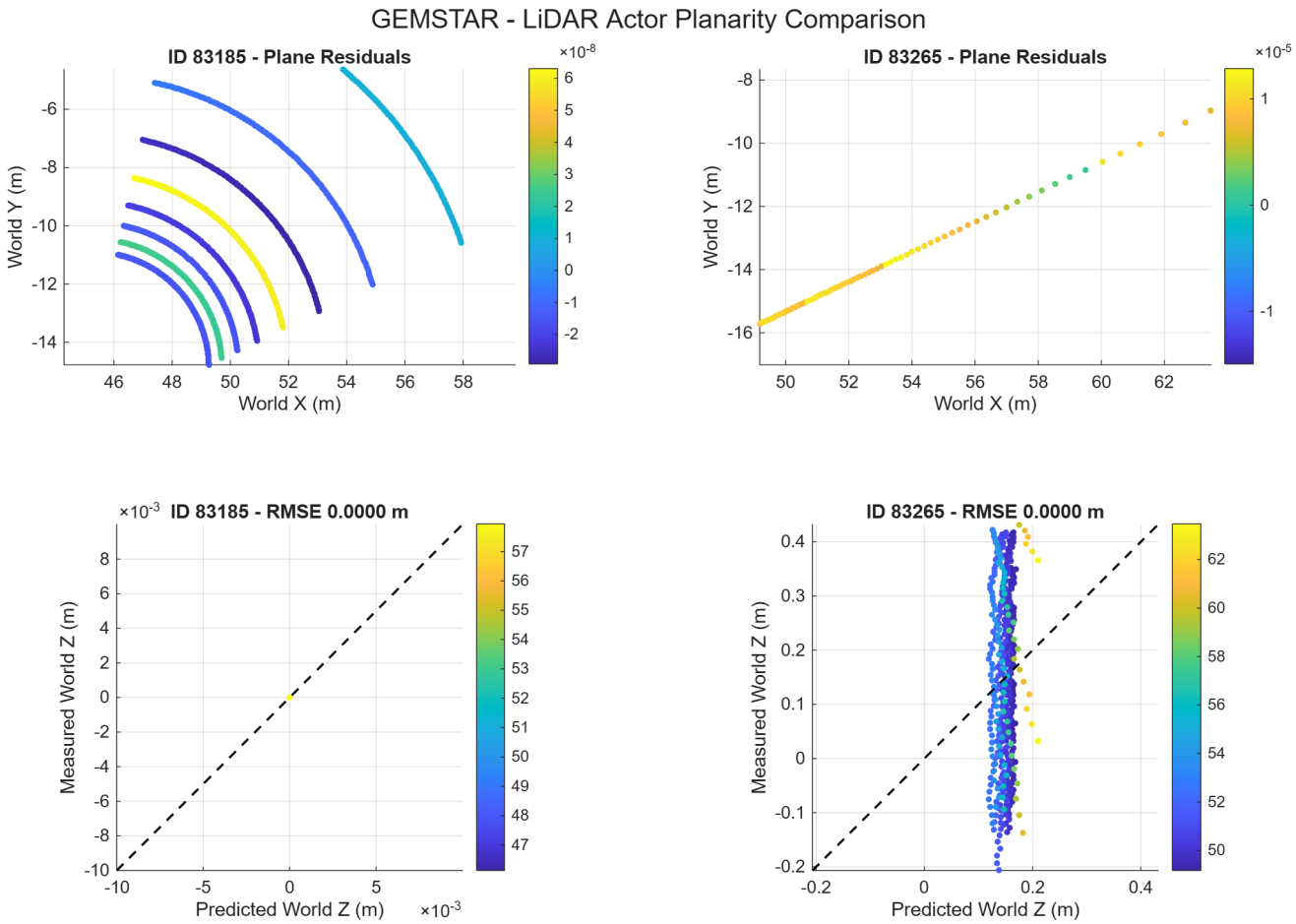
actor geometry comparison



Original saved graph: 01_Graphs\task19_8k_actor_geometry_comparison.png

Research Figure 39

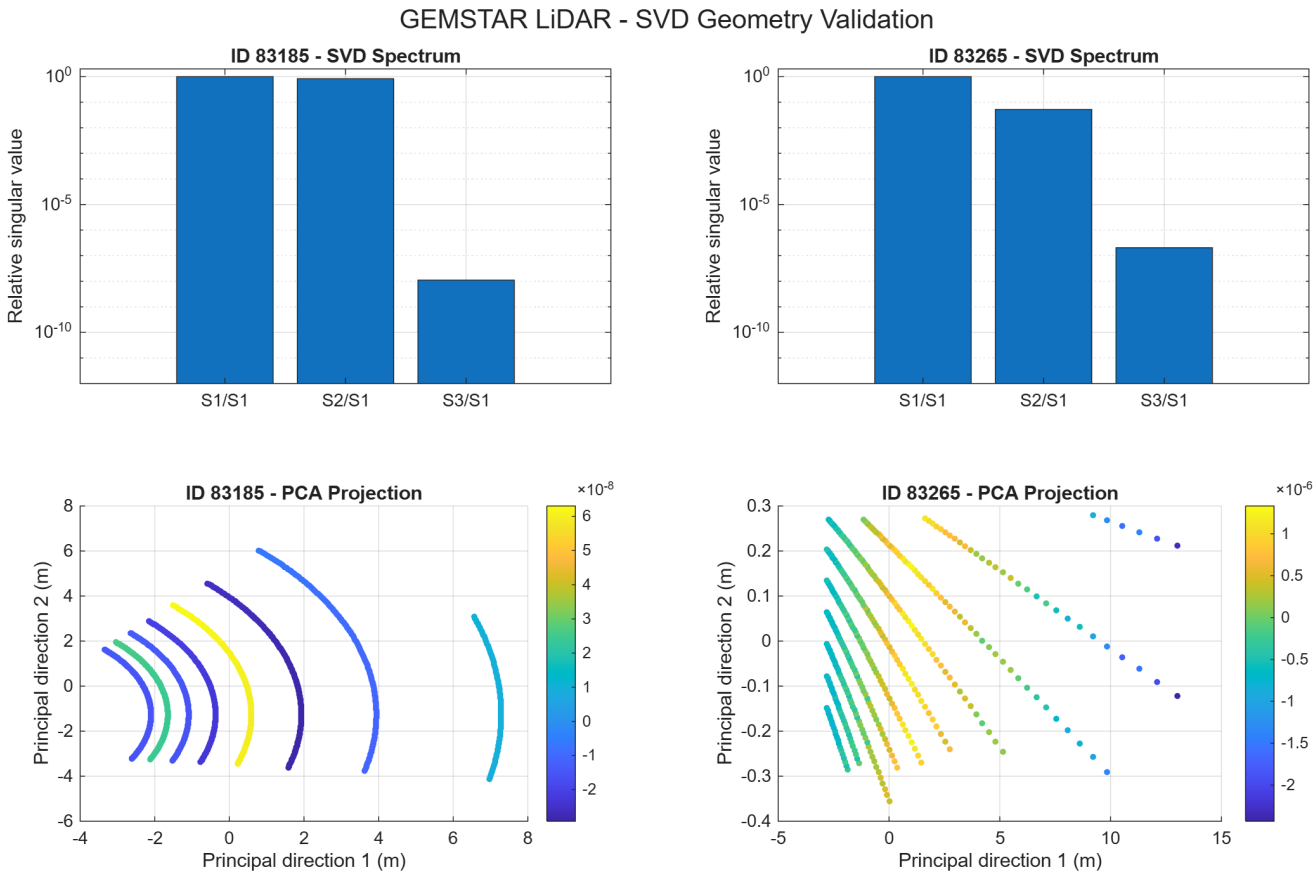
actor planarity analysis



Original saved graph: 01_Graphs\task19_8l_actor_planarity_analysis.png

Research Figure 40

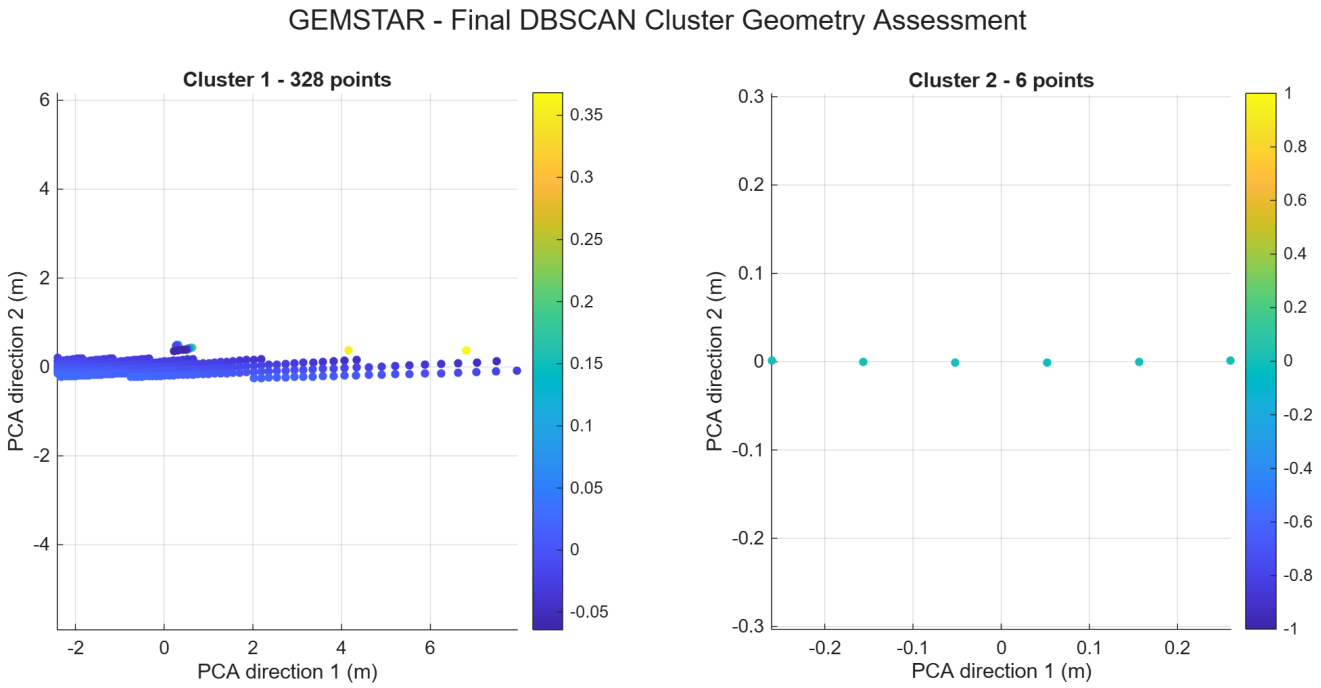
svd geometry



Original saved graph: 01_Graphs\task19_8m_svd_geometry.png

Research Figure 41

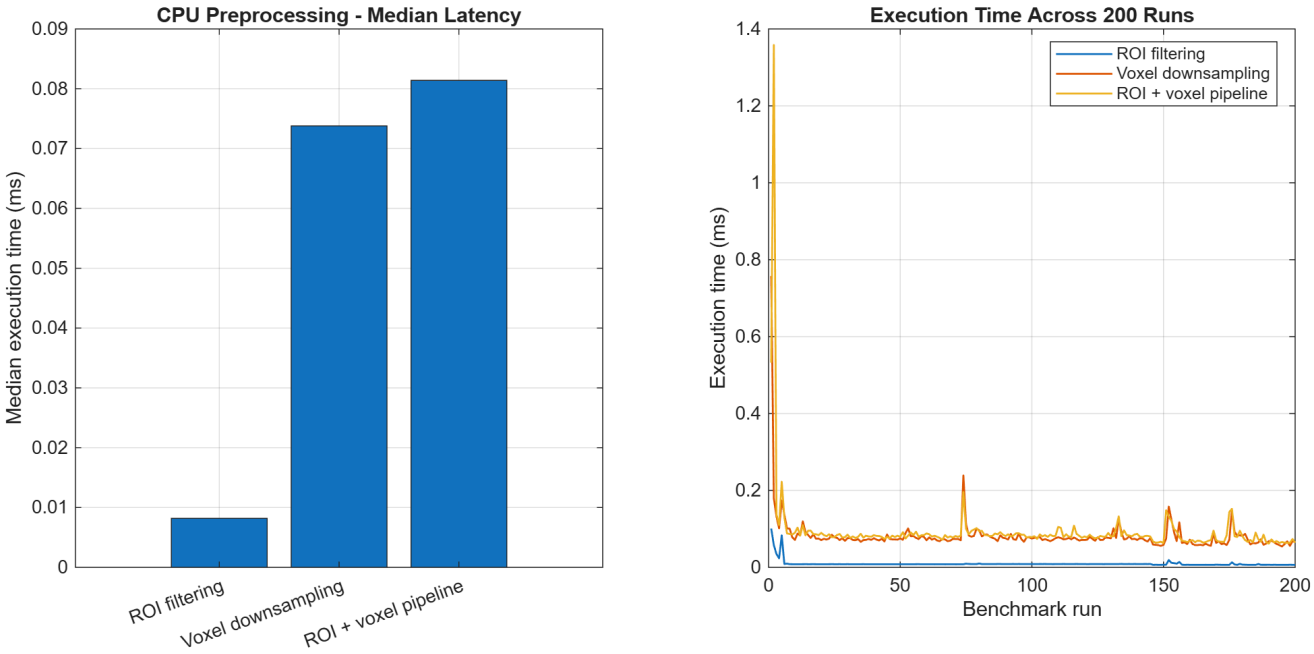
cluster quality



Original saved graph: 01_Graphs\task19_8n_cluster_quality.png

Research Figure 42

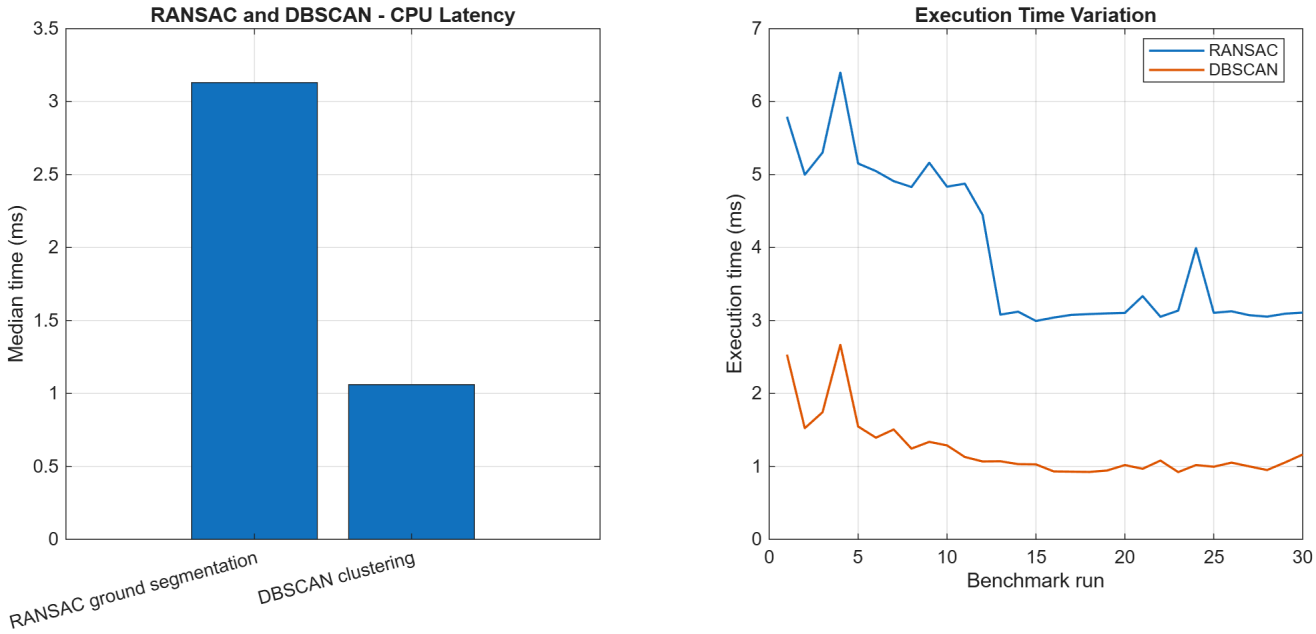
cpu latency benchmark



Original saved graph: 01_Graphs\task19_9a_cpu_latency_benchmark.png

Research Figure 43

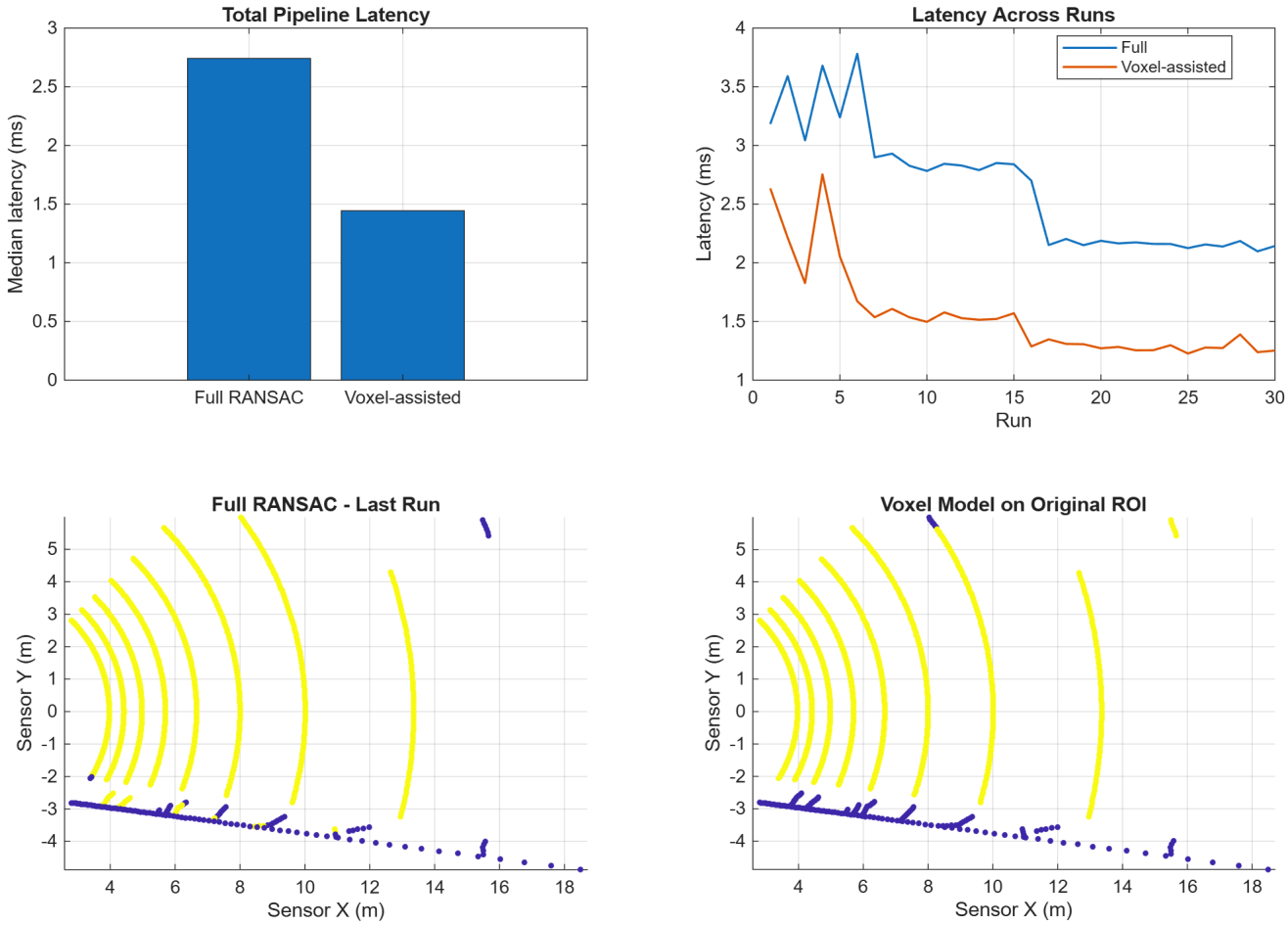
ransac dbscan latency



Original saved graph: 01_Graphs\task19_9b_ransac_dbscan_latency.png

Research Figure 44

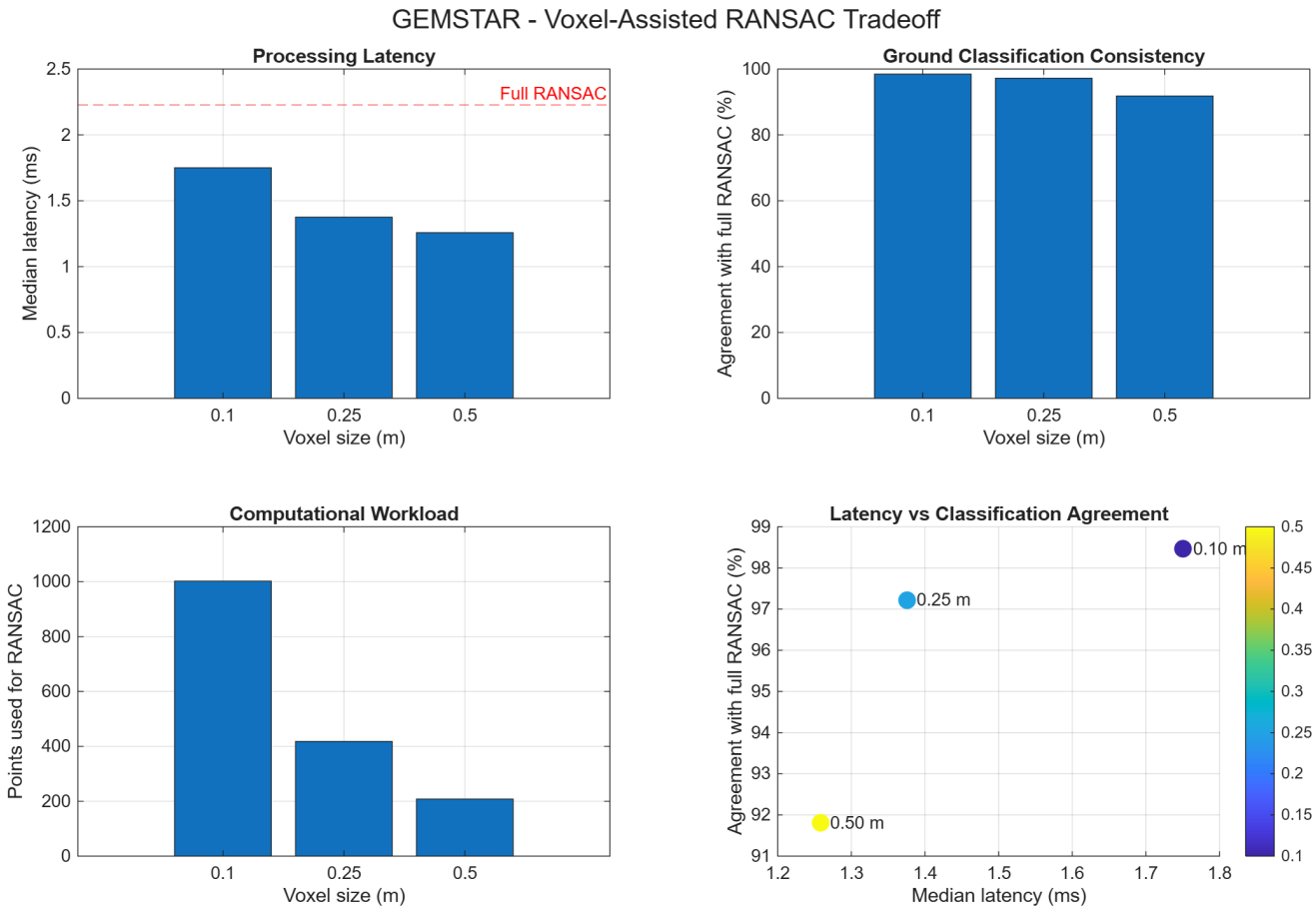
voxel ransac tradeoff



Original saved graph: 01_Graphs\task19_9c_voxel_ransac_tradeoff.png

Research Figure 45

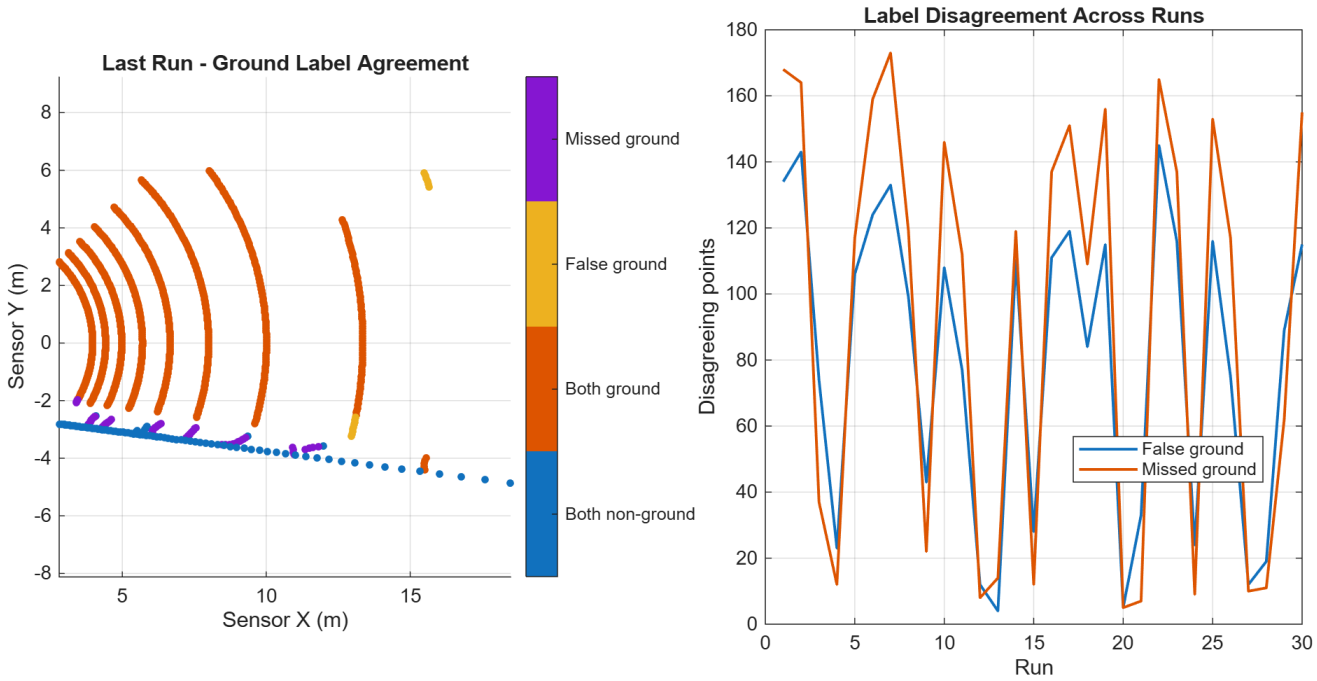
voxel size tradeoff



Original saved graph: 01_Graphs\task19_9d_voxel_size_tradeoff.png

Research Figure 46

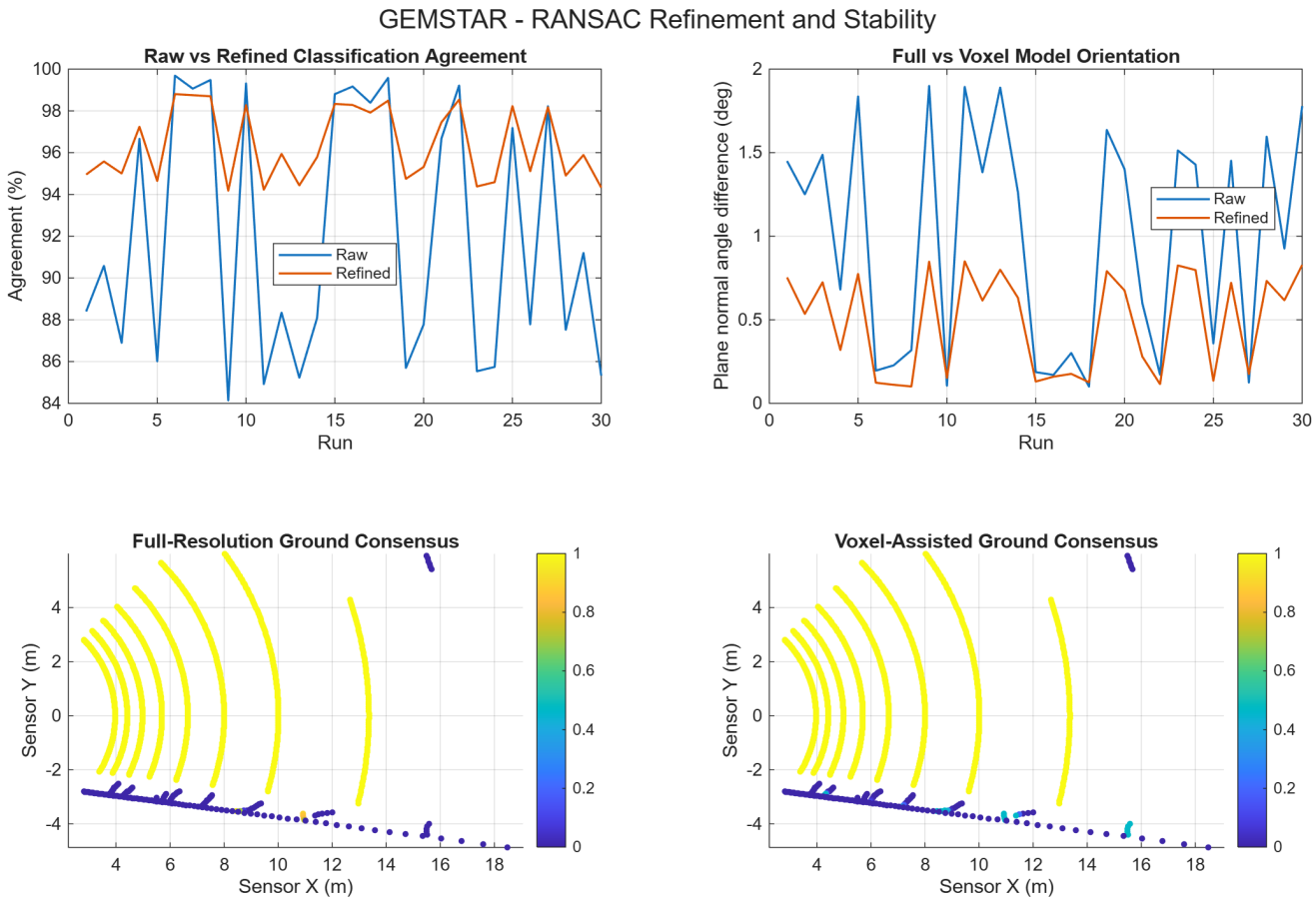
ground label risk



Original saved graph: 01_Graphs\task19_9e_ground_label_risk.png

Research Figure 47

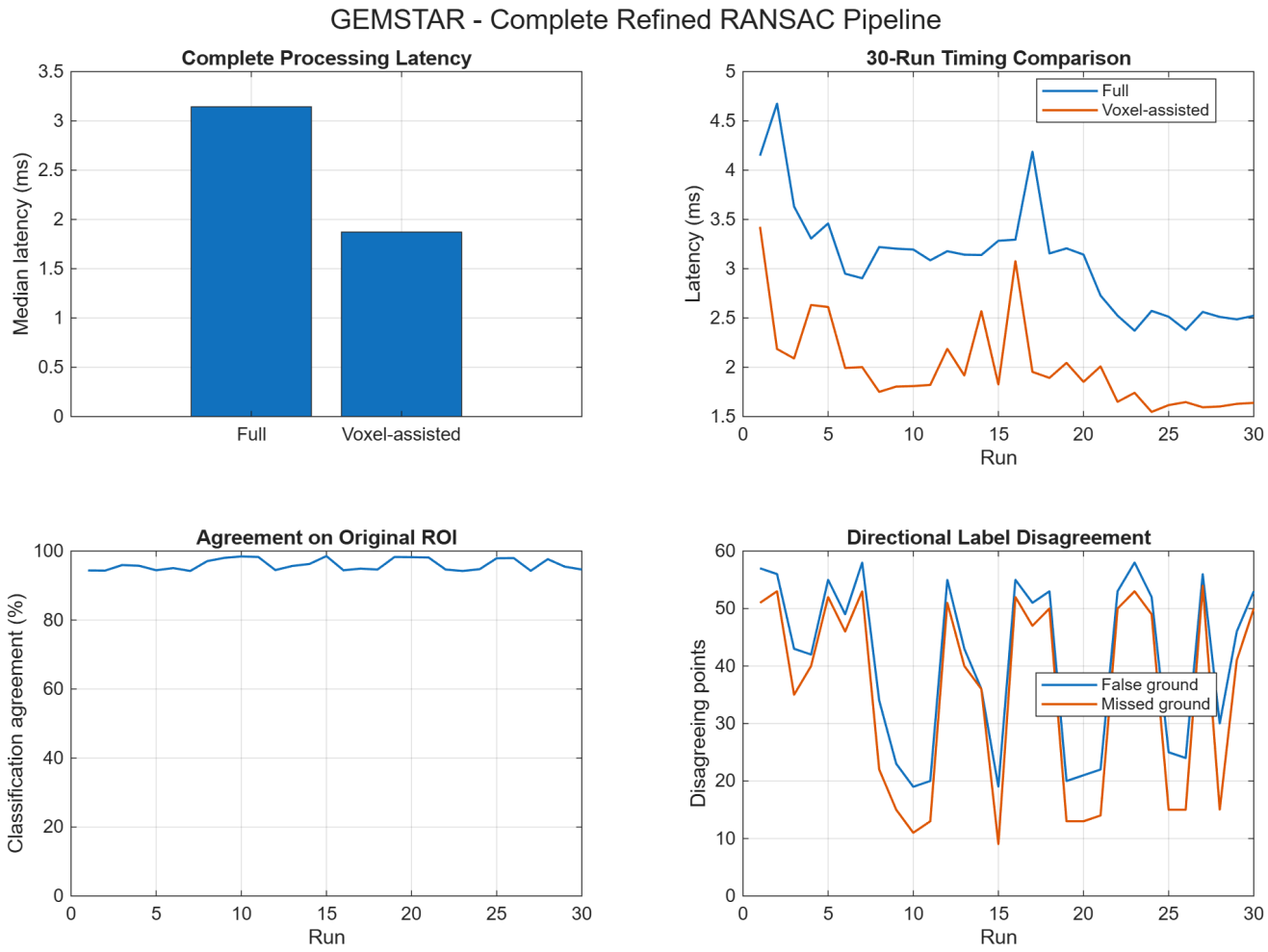
ransac refinement stability



Original saved graph: 01_Graphs\task19_9f_ransac_refinement_stability.png

Research Figure 48

refined pipeline benchmark



Original saved graph: 01_Graphs\task19_9g_refined_pipeline_benchmark.png

14. Reproducibility and Research Archive

Archive section	Manifest files
01_Graphs	48
02_CSV_Data	85
03_Configurations	1
04_Code	56
05_Raw_Data_References	9
06_Research_Findings	1
08_Extracted_PointClouds	86

Original manifest verified: 286 verified, 0 missing, 0 changed.

The original manifest is a point-in-time snapshot created in Task 19.15. Files generated after that snapshot, including this final PDF, are intentionally reported separately.

Full file hashes and verification statuses are saved in 06_Research_Findings. Source code and raw-data references remain in their existing archive folders.